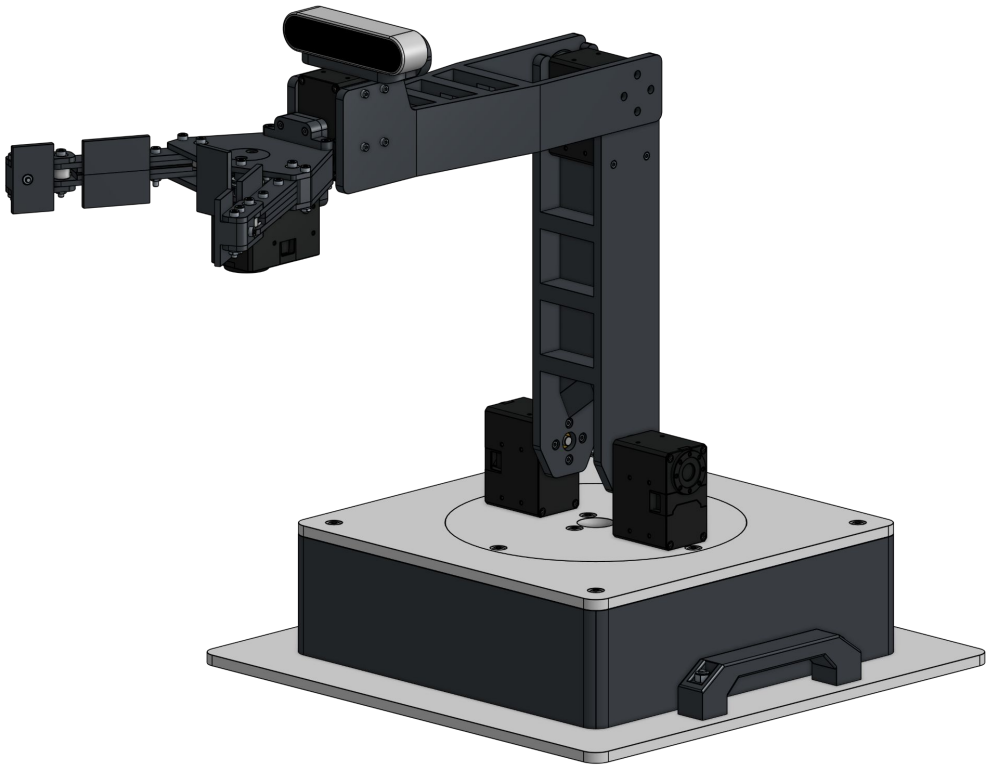
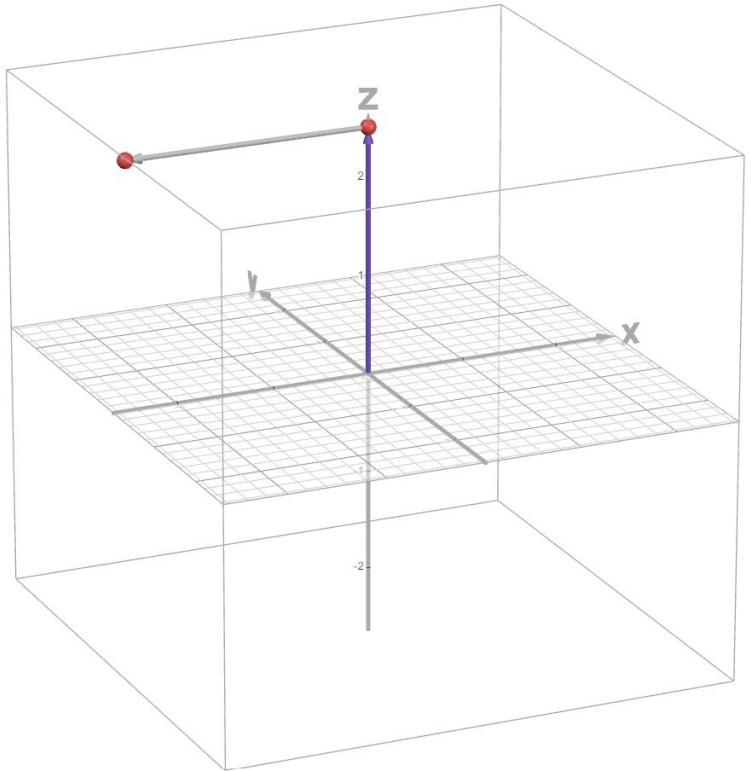
The Trigonometric Approach to Forward Kinematics

By Joseph Ruhan





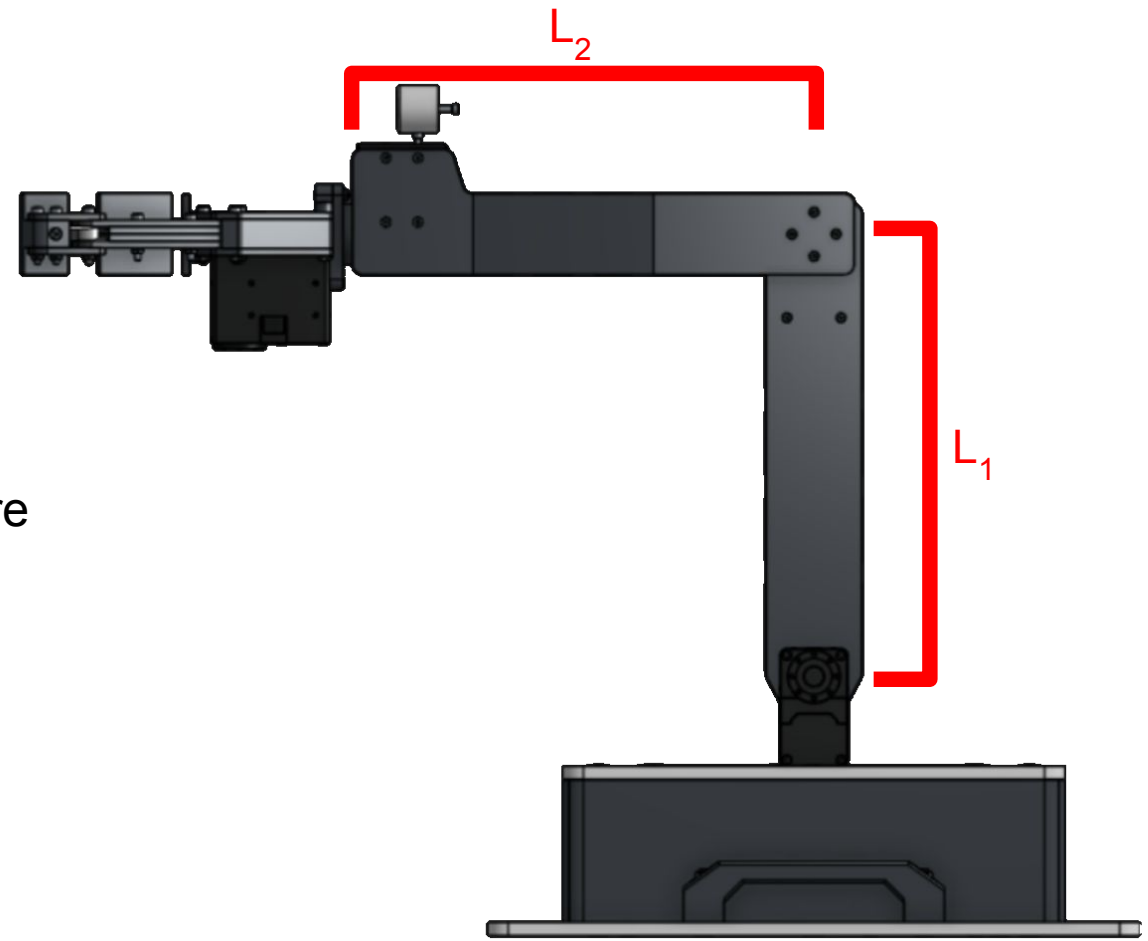
The arm can be described mathematically by defining its joints and links

Where

L_1=250mm

L_2=250mm

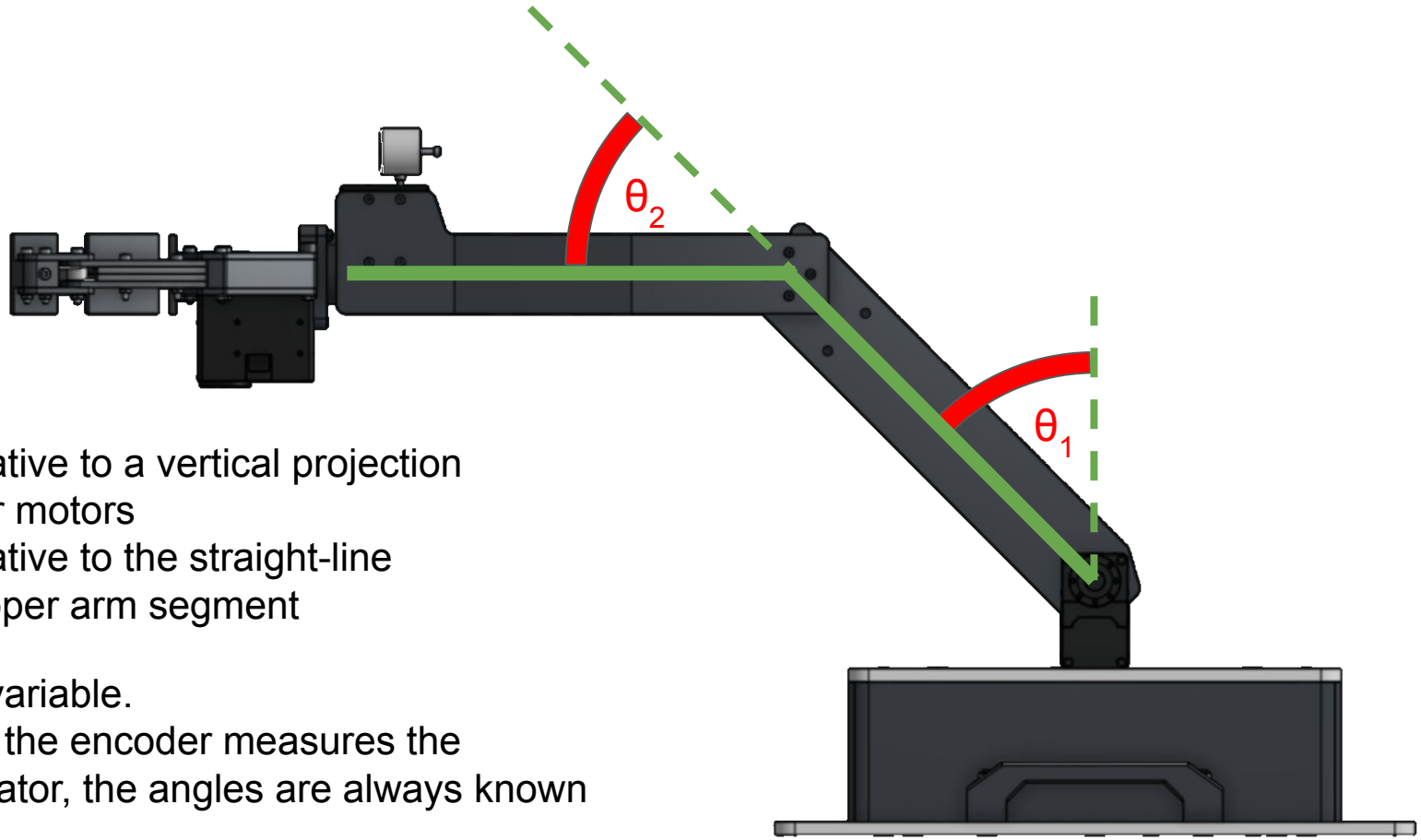
These are always known because they are a constant



In this example

θ_1=45°

θ_2=45°



θ_1 is measured relative to a vertical projection above the shoulder motors

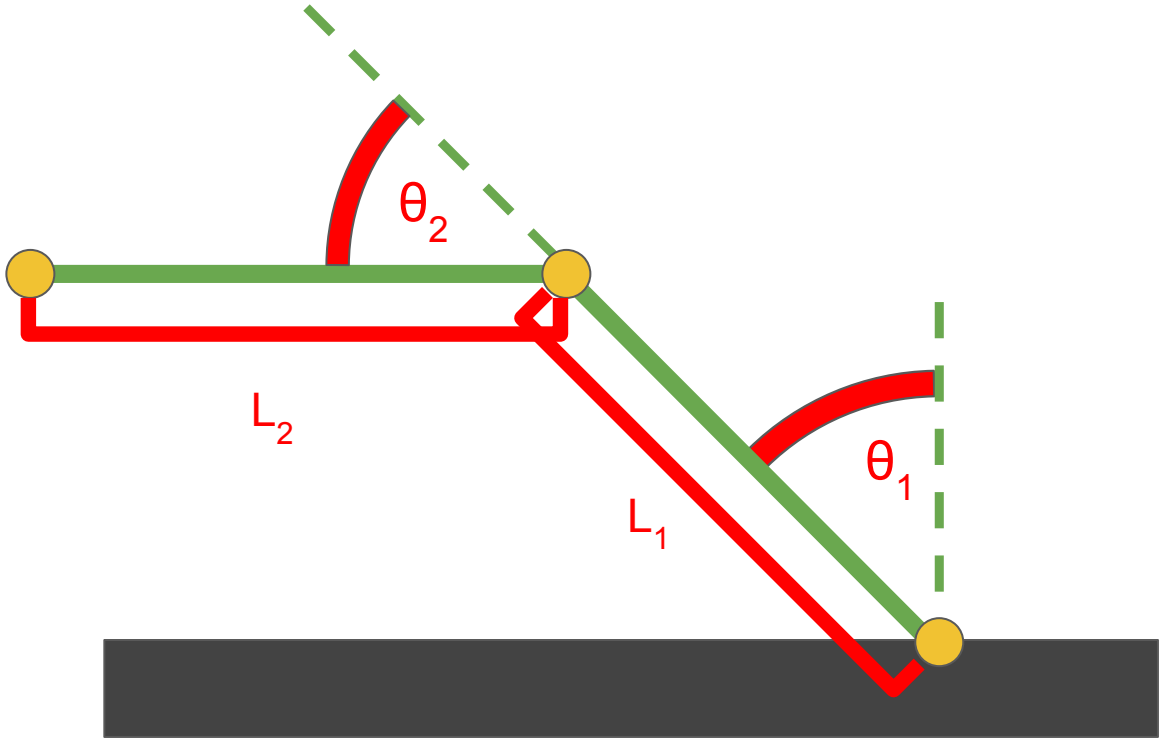
θ_2 is measured relative to the straight-line extension of the upper arm segment

These angles are variable.
However, because the encoder measures the rotation of the actuator, the angles are always known

The arm can therefore be represented as such:

= a joint

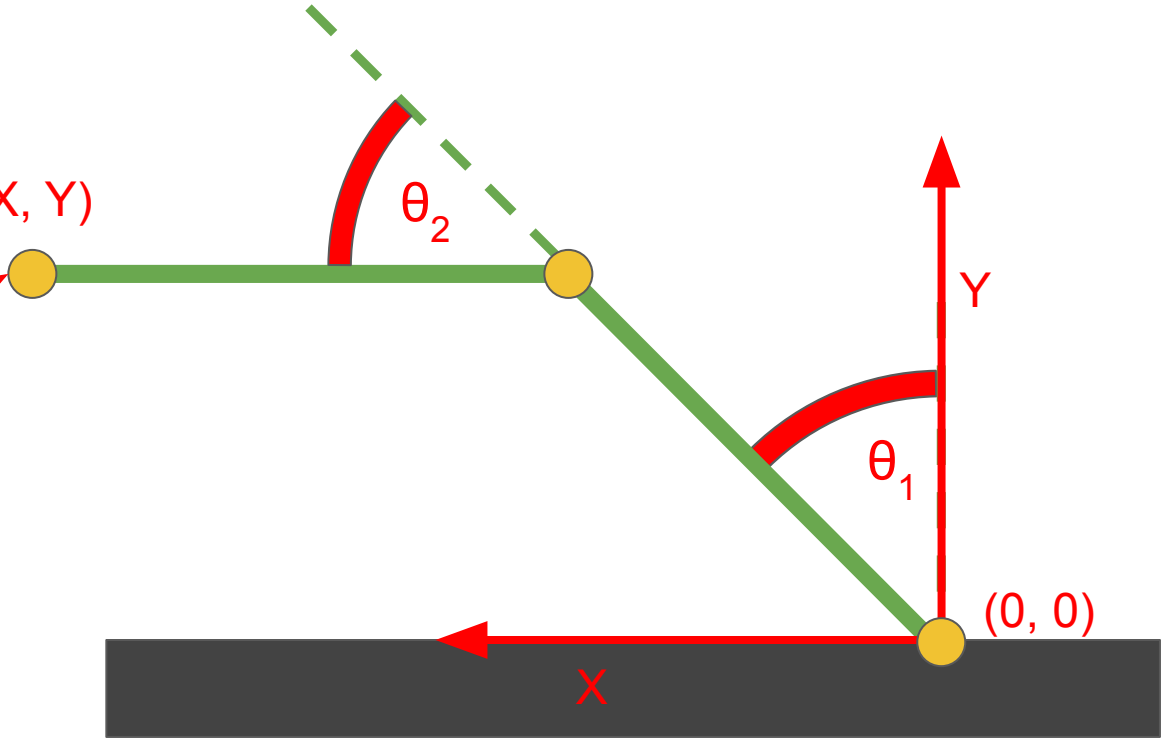
L_1=250mm
L_2=250mm
θ_1=45°
θ_2=45°



The arm can therefore be represented as such:

= a joint

L_1=250mm
L_2=250mm
θ_1=45°
θ_2=45°

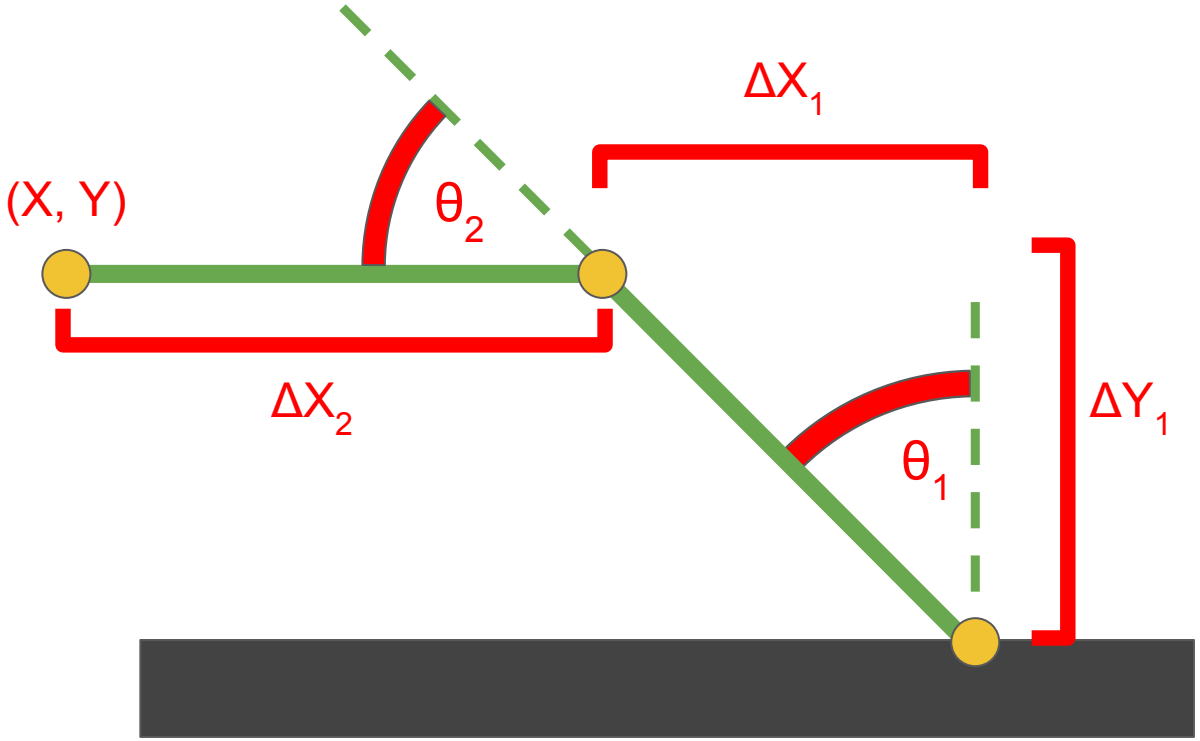


The location of the end effector is the unknown variable

The arm can therefore be represented as such:

= a joint

L_1=250mm
L_2=250mm
θ_1=45°
θ_2=45°

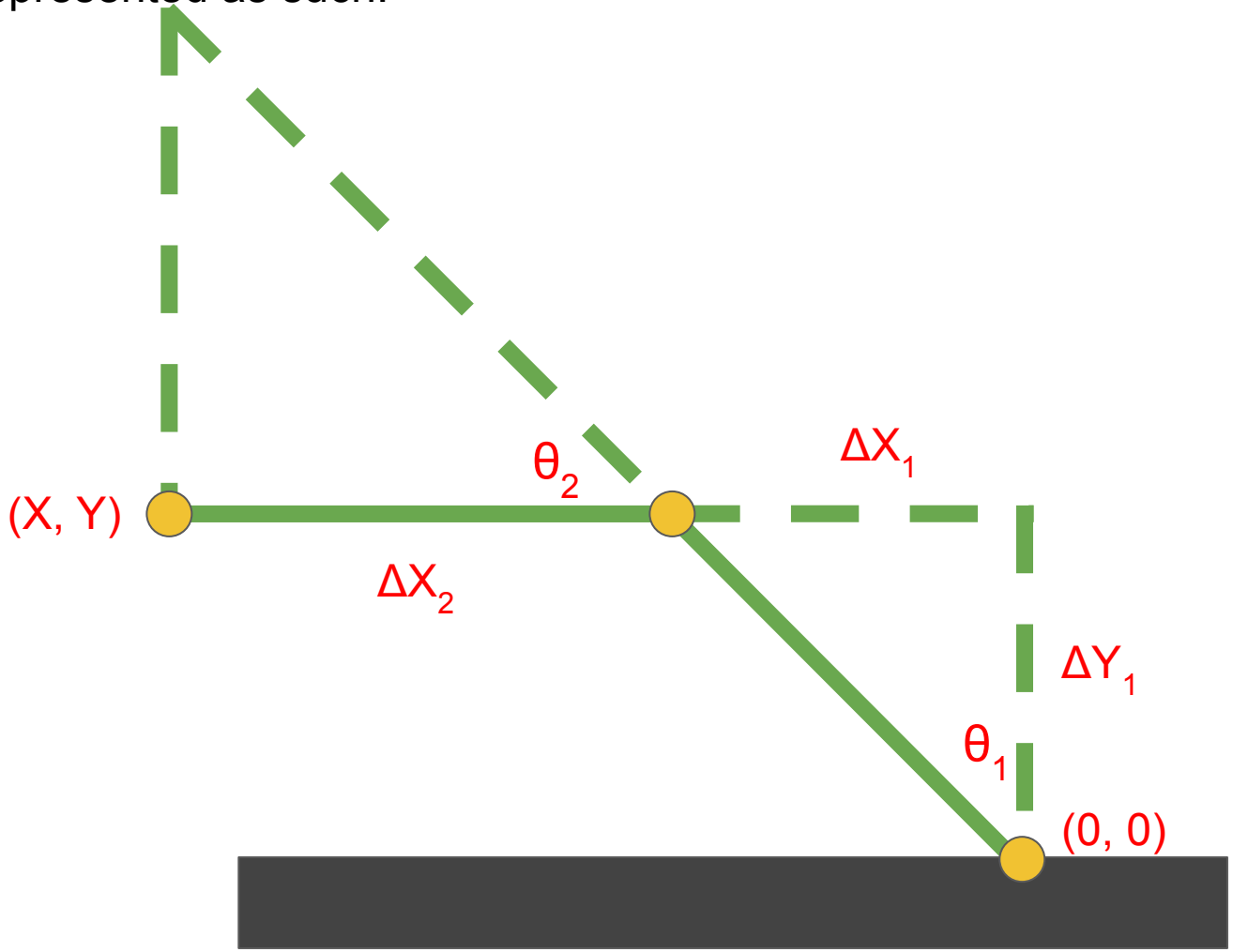


The position of the end effector will be a sum of the vertical and horizontal displacement of each joint

The arm can therefore be represented as such:

= a joint

L_1=250mm
L_2=250mm
θ_1=45°
θ_2=45°



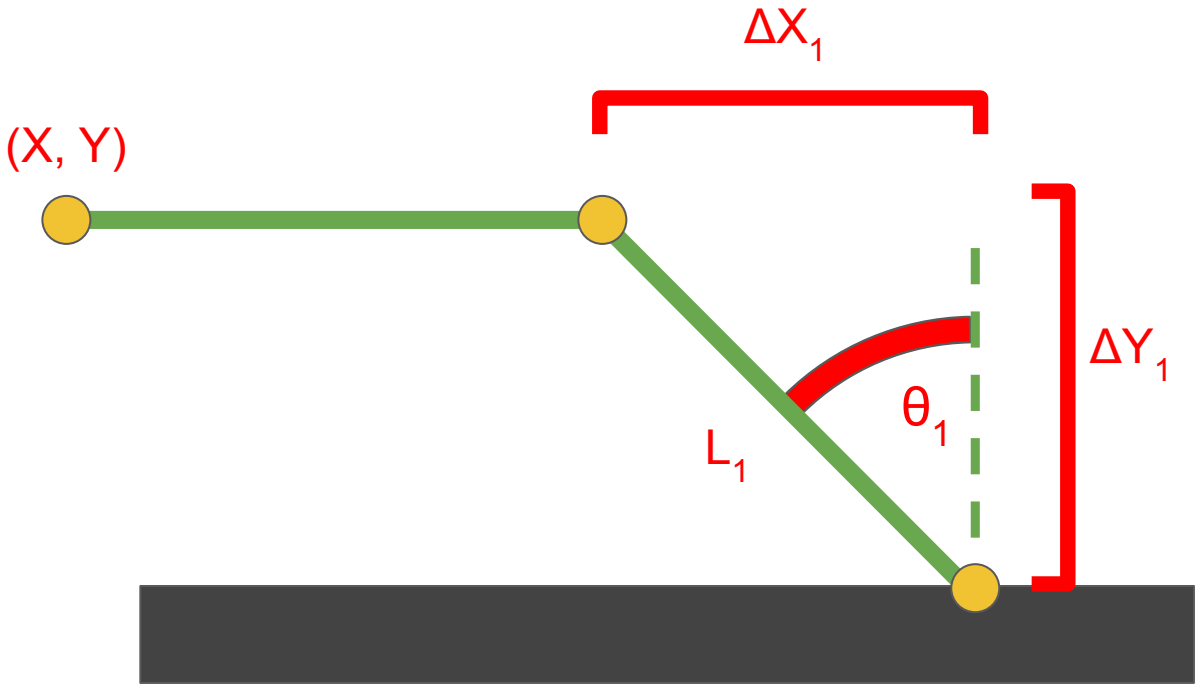
The displacement can be represented by triangles

Each link has its own triangle

L_1 is the hypotenuse of the upper arm triangle (ΔX_1, ΔY_1)

To use sin and cos, we need the angle between the link and vertical

Upper arm: θ_1 is already measured from vertical, so we use θ_1



The forearm uses $\theta_1 + \theta_2$

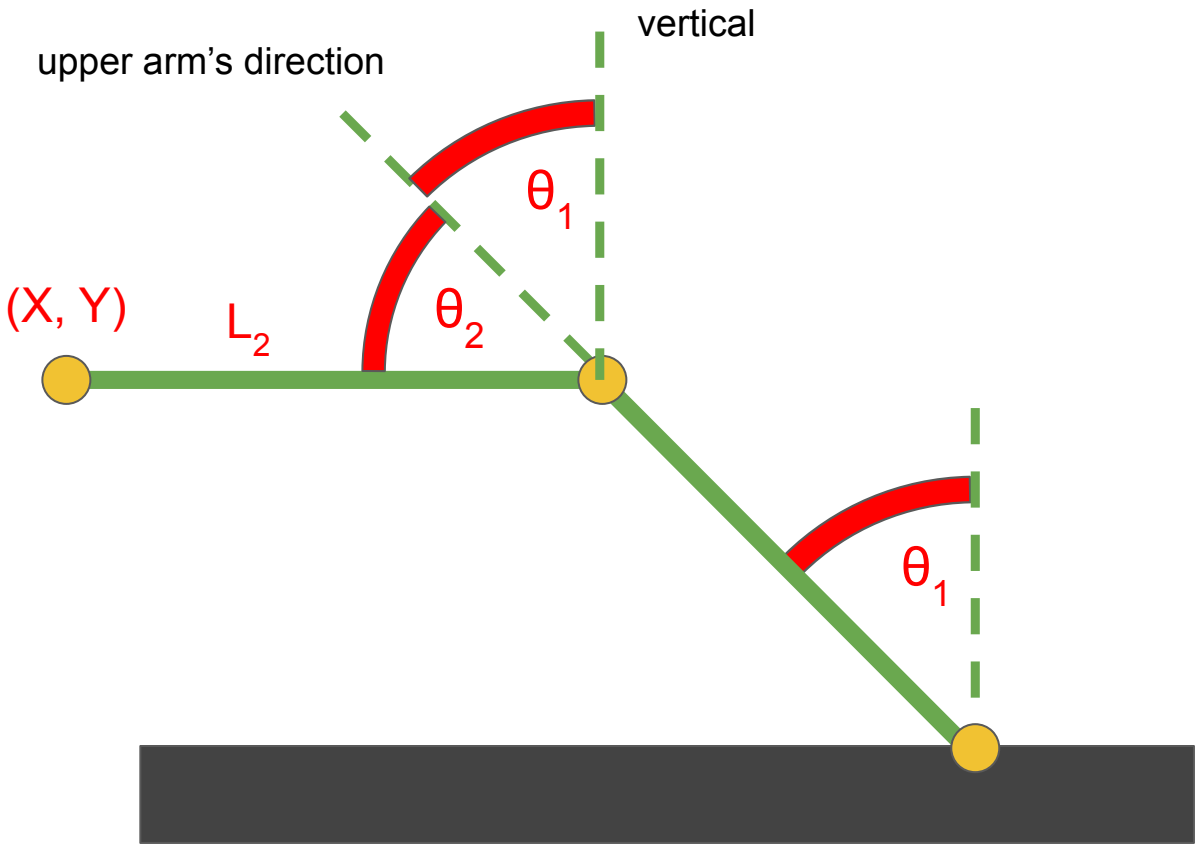
L_2 is the hypotenuse of the forearm triangle (ΔX_2, ΔY_2)

θ_2 is measured from the upper arm's direction, not from vertical

From vertical, the forearm has turned $\theta_1 + \theta_2$

Here: 45° + 45° = 90°, so the forearm is horizontal

Check: if θ_2 = 0° the arm is straight, so the angle is just θ_1



To calculate the position of the end point:

L_1=250mm
L_2=250mm
θ_1=45°
θ_2=45°

In this example, L_1 is the hypotenuse of a triangle.

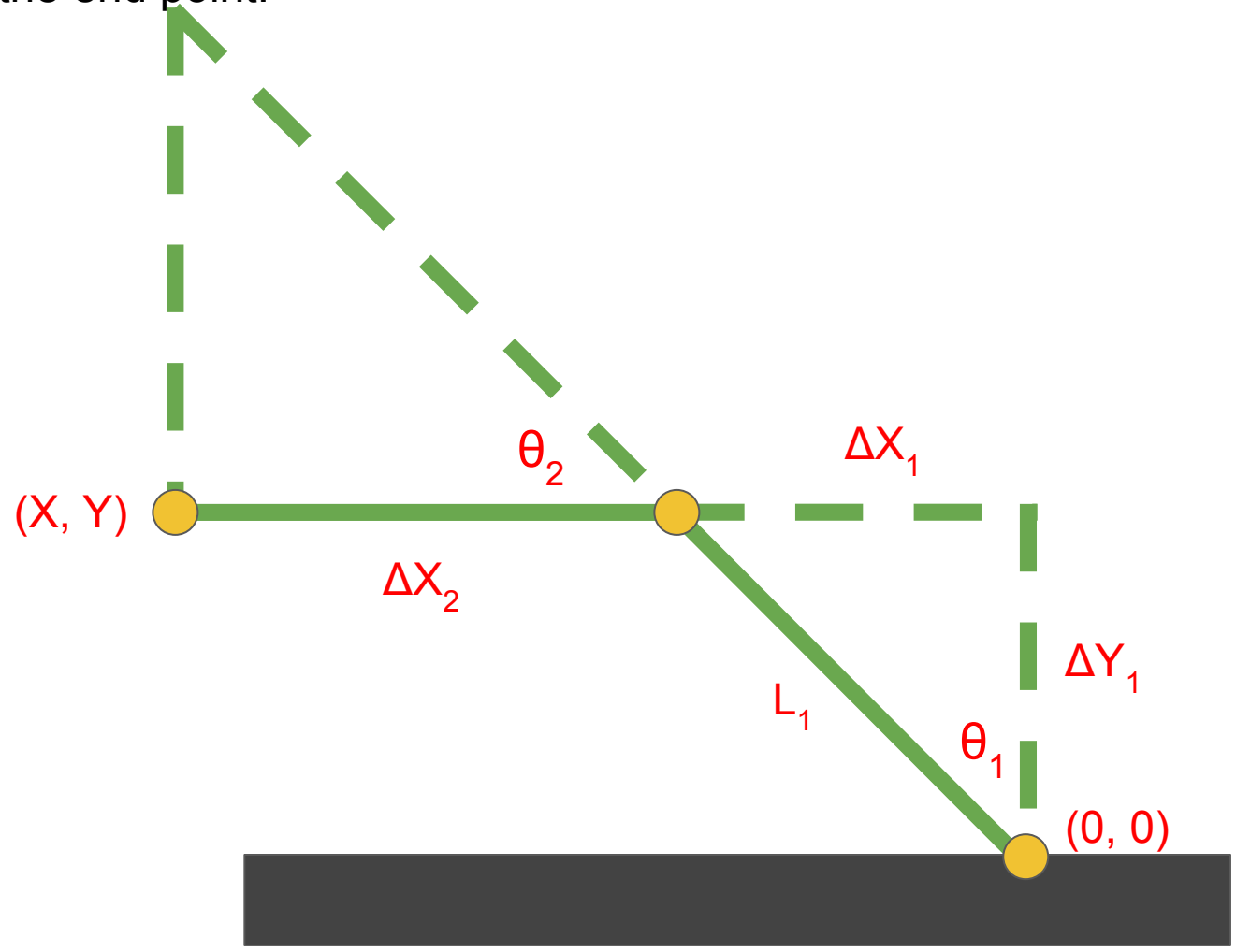
Using $\sin(\theta_1)$:

$\sin(\theta_1) = \Delta X_1 / L_1$

$L_1 \sin(\theta_1) = \Delta X_1$

$250*(\sqrt{2}/2) = \Delta X_1$

$\Delta X_1 = 176.777mm$



To calculate the position of the end point:

$\Delta X_1 = 176.777mm$

For $\cos(\theta_1)$:

$\cos(\theta_1) = \Delta Y_1 / L_1$

$L_1\cos(\theta_1) = \Delta Y_1$

$250*(\sqrt{2}/2) = \Delta Y_1$

$\Delta Y_1 = 176.777mm$

The forearm angle is $\theta_1+\theta_2 = 90°$

$\Delta X_2 = L_2\sin(\theta_1+\theta_2)$

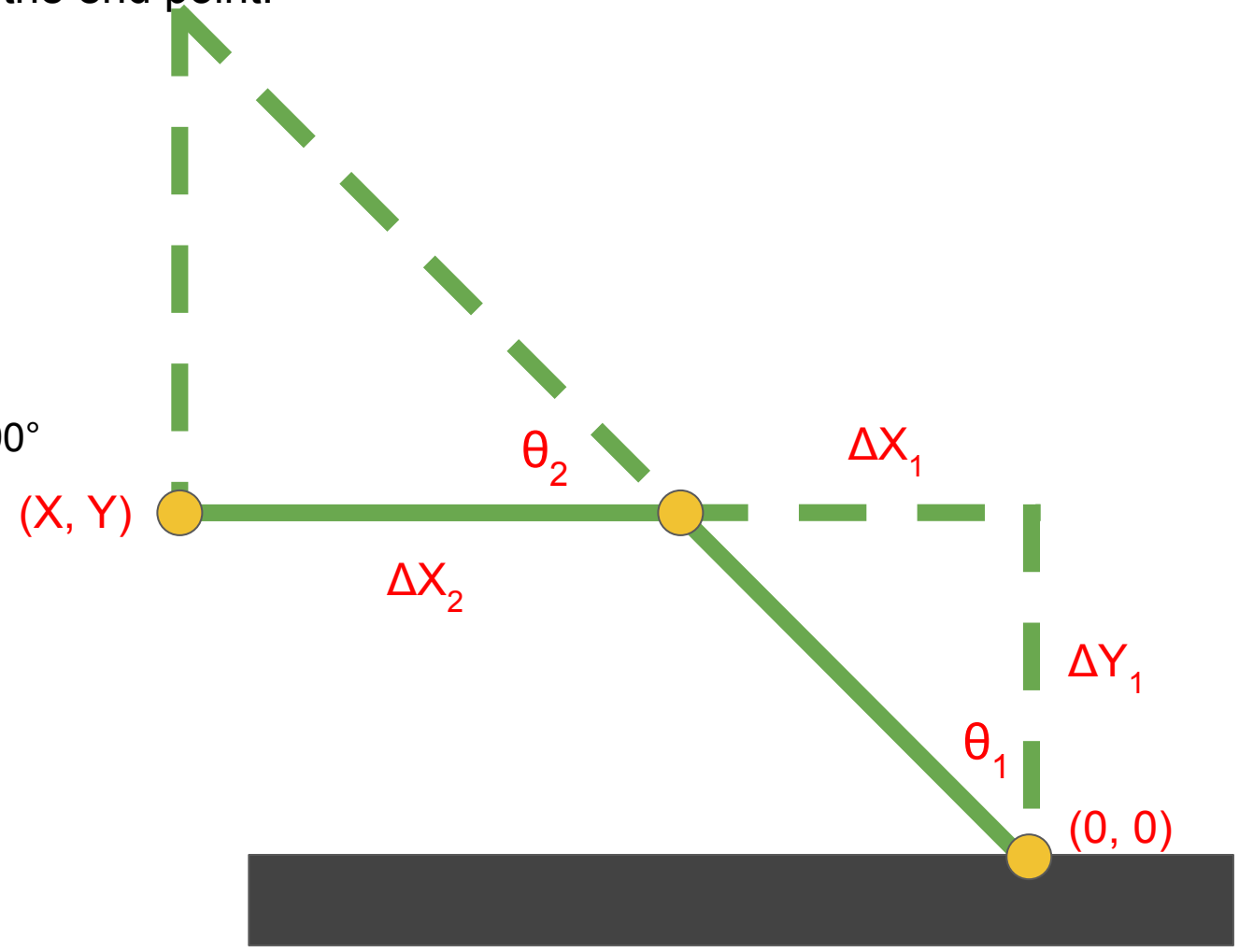
$250*\sin(90°) = \Delta X_2$

$\Delta X_2 = 250mm$

$\Delta Y_2 = L_2\cos(\theta_1+\theta_2)$

$250*\cos(90°) = \Delta Y_2$

$\Delta Y_2 = 0mm$



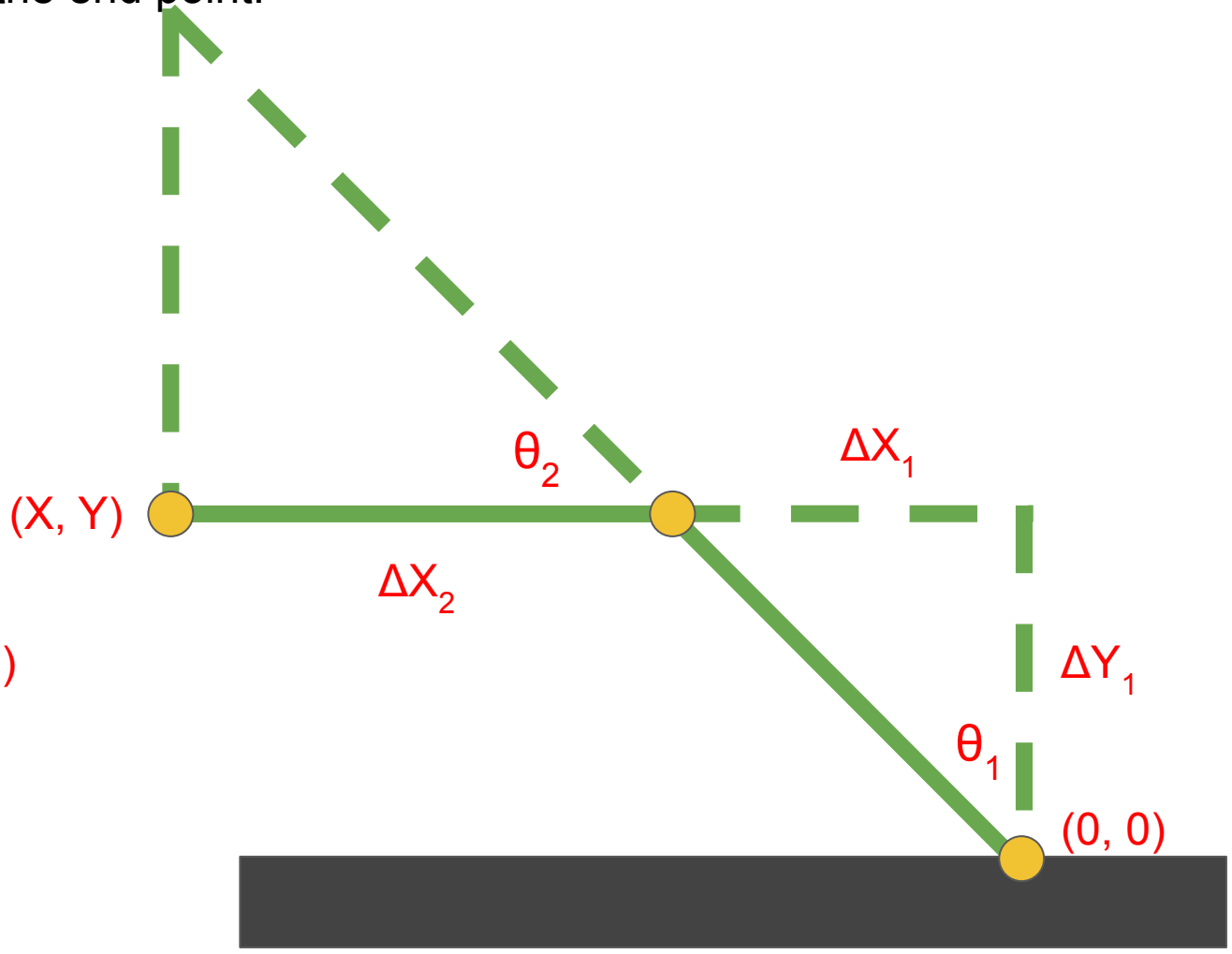
To calculate the position of the end point:

ΔX_1 = 176.777mm

ΔY_1 = 176.777mm

ΔX_2 = 250mm

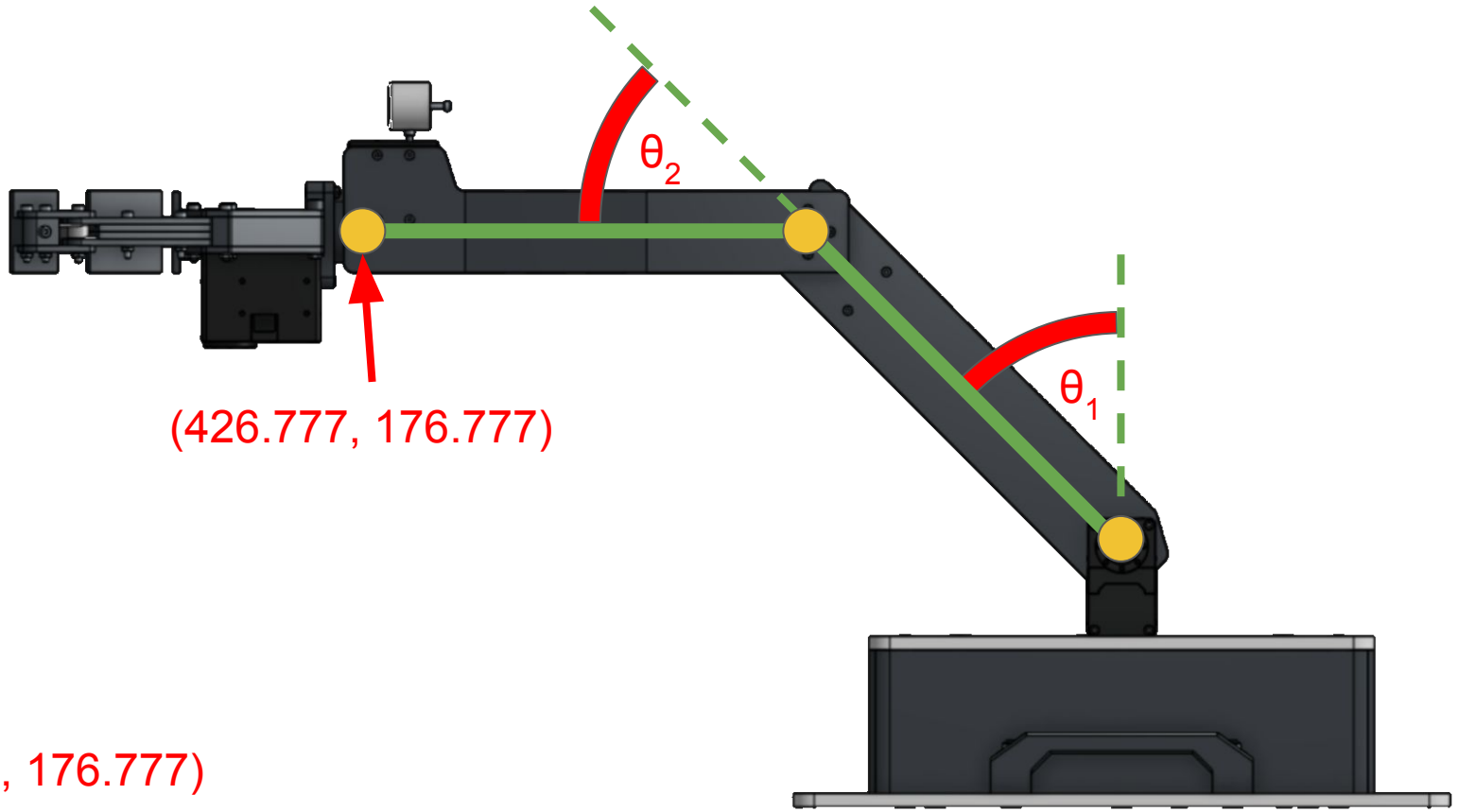
ΔY_2 = 0mm



$(X,Y) = (\Delta X_1 + \Delta X_2, \Delta Y_1 + \Delta Y_2)$

(X,Y) = (426.777, 176.777)

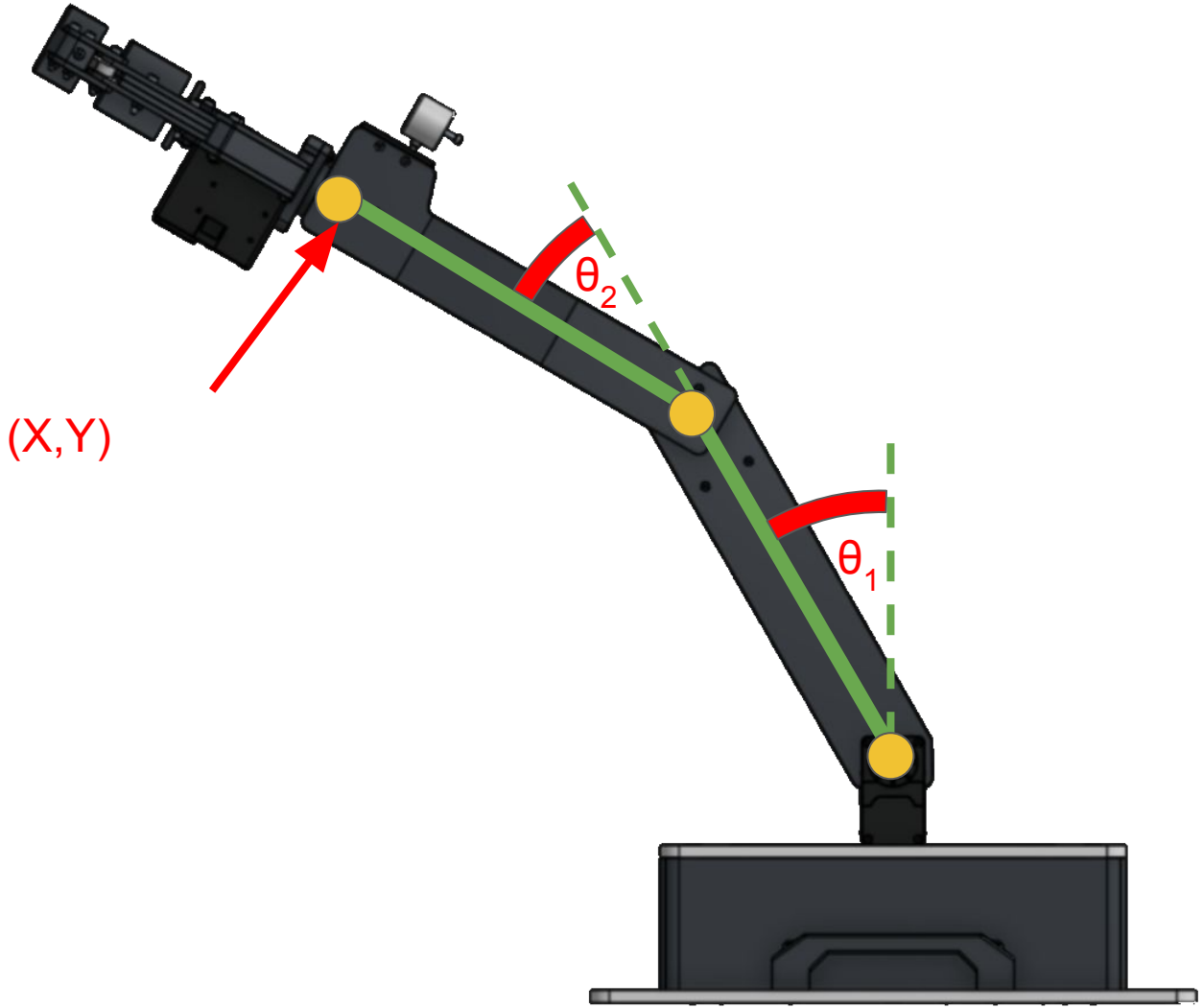
Completed Variables



L_1=250mm
L_2=250mm
θ_1=45°
θ_2=45°
(X,Y) = (426.777, 176.777)

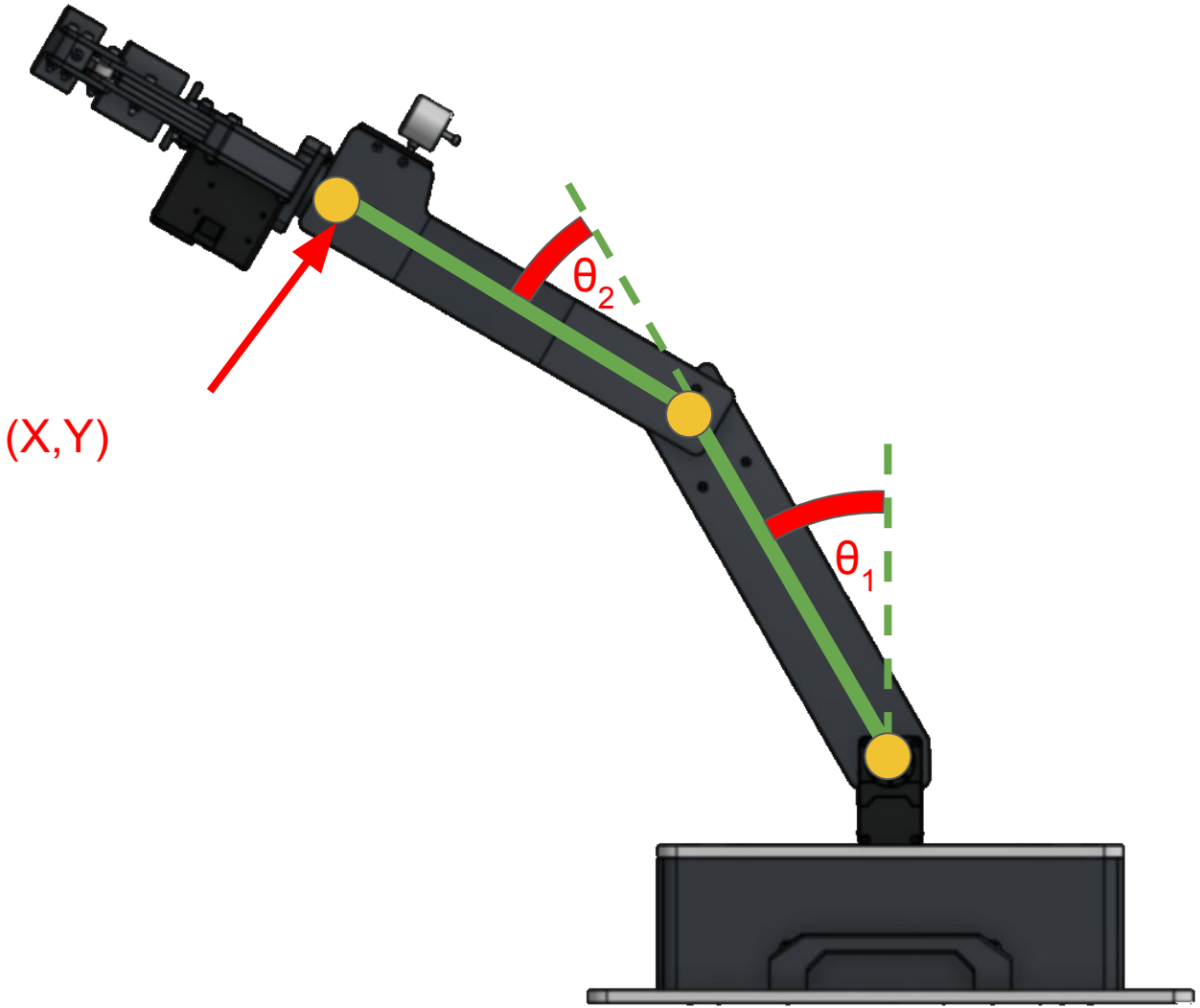
Try It Yourself:

L_1=250mm
L_2=250mm
θ_1=30°
θ_2=30°
(X,Y) = (?,?)



Try It Yourself:

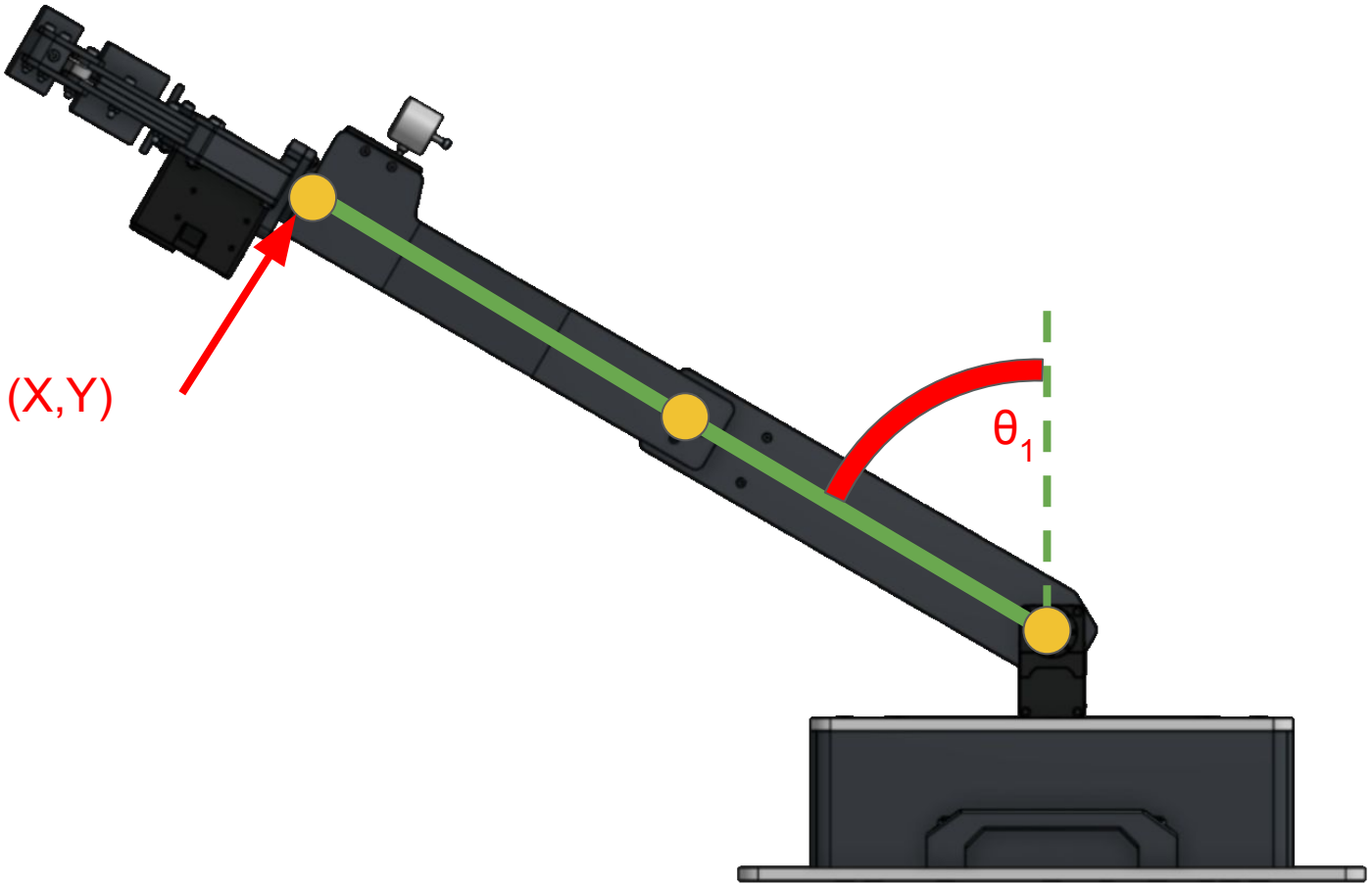
L_1=250mm
L_2=250mm
θ_1=30°
θ_2=30°
(X,Y) = (?,?)



Solution:
(X,Y) = (341.506, 341.506)

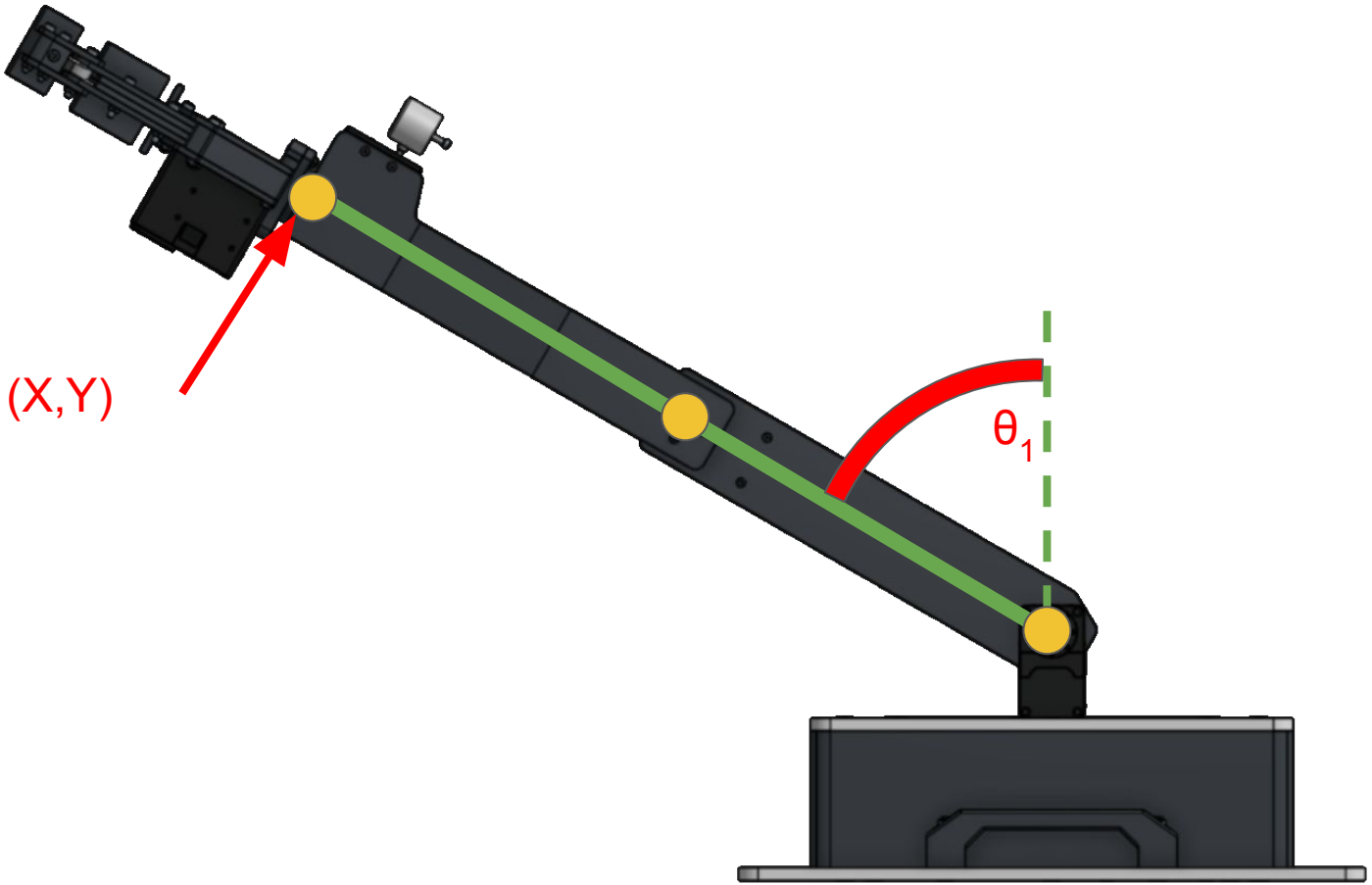
Try It Yourself 2:

L_1=250mm
L_2=250mm
θ_1=60°
θ_2=0°
(X,Y) = (?,?)



Try It Yourself 2:

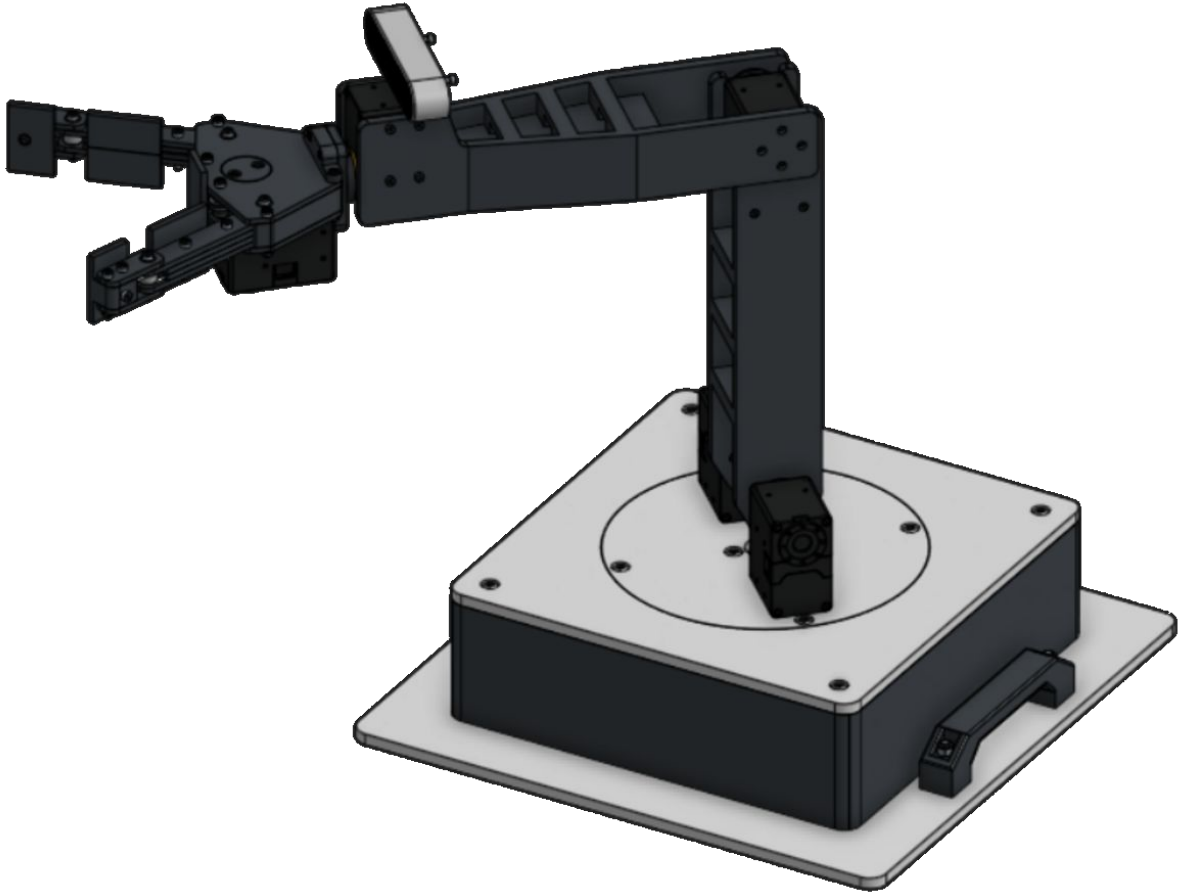
L_1=250mm
L_2=250mm
θ_1=60°
θ_2=0°
(X,Y) = (?,?)



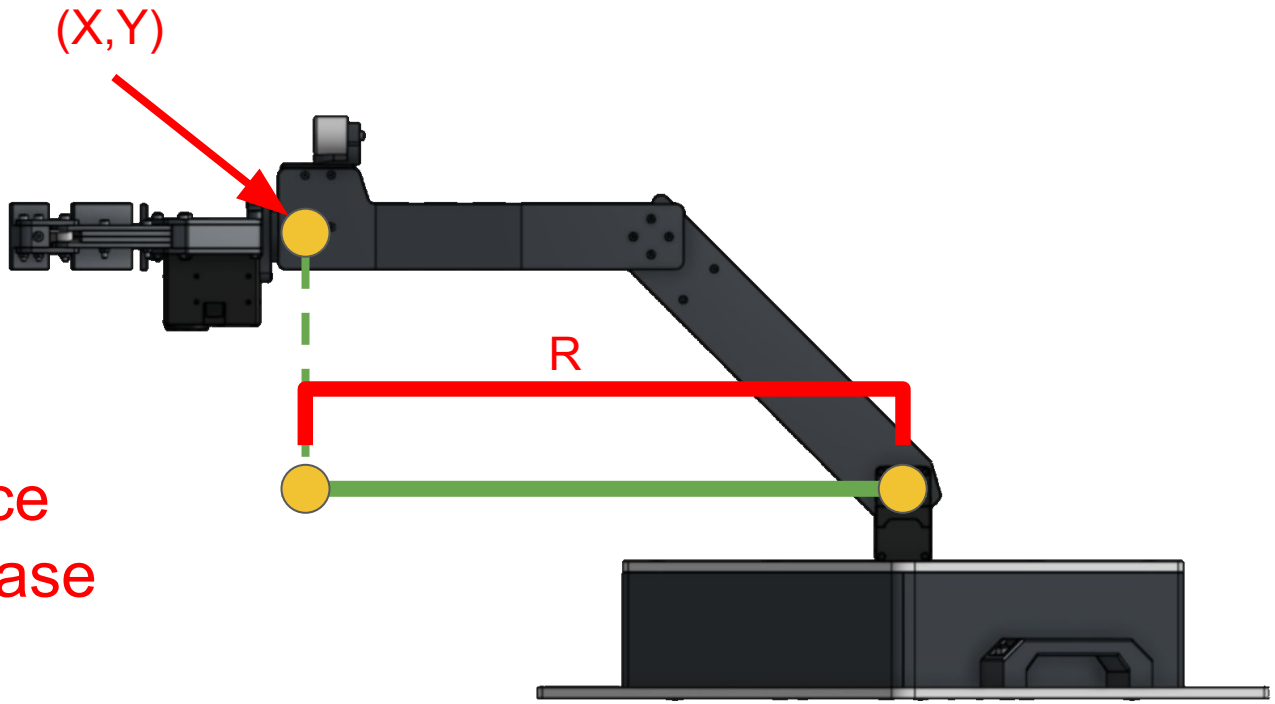
Solution:

(X,Y) = (433.013, 250)

Forward Kinematics With Base Rotation



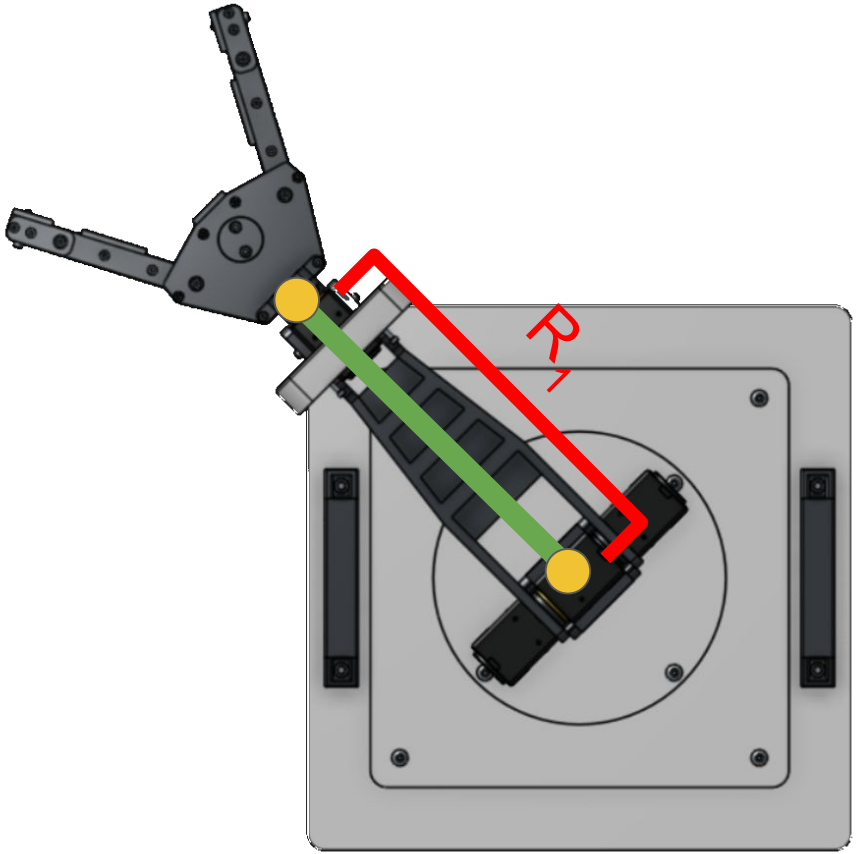
The arm's forward kinematics with base rotation is dependent on rotation around the center point of the base



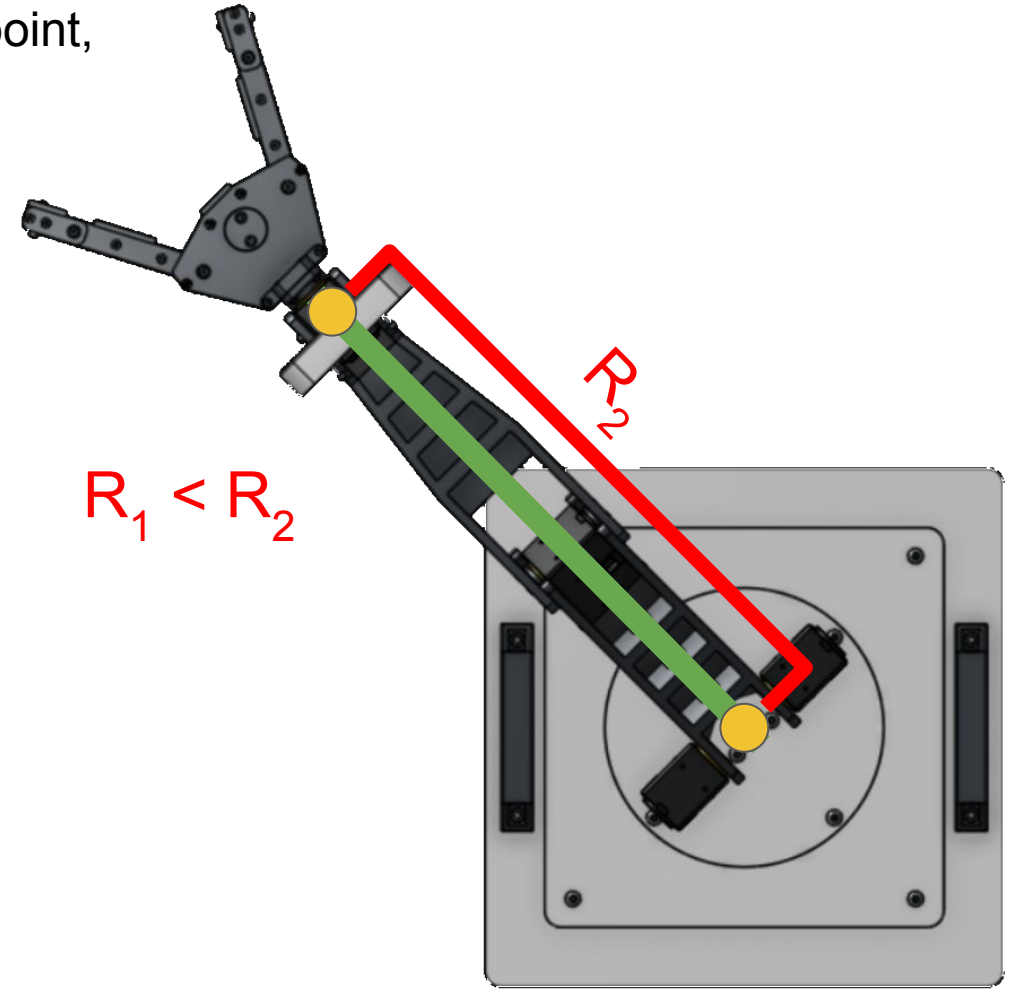
Where

R = Horizontal distance from end effector to base center point

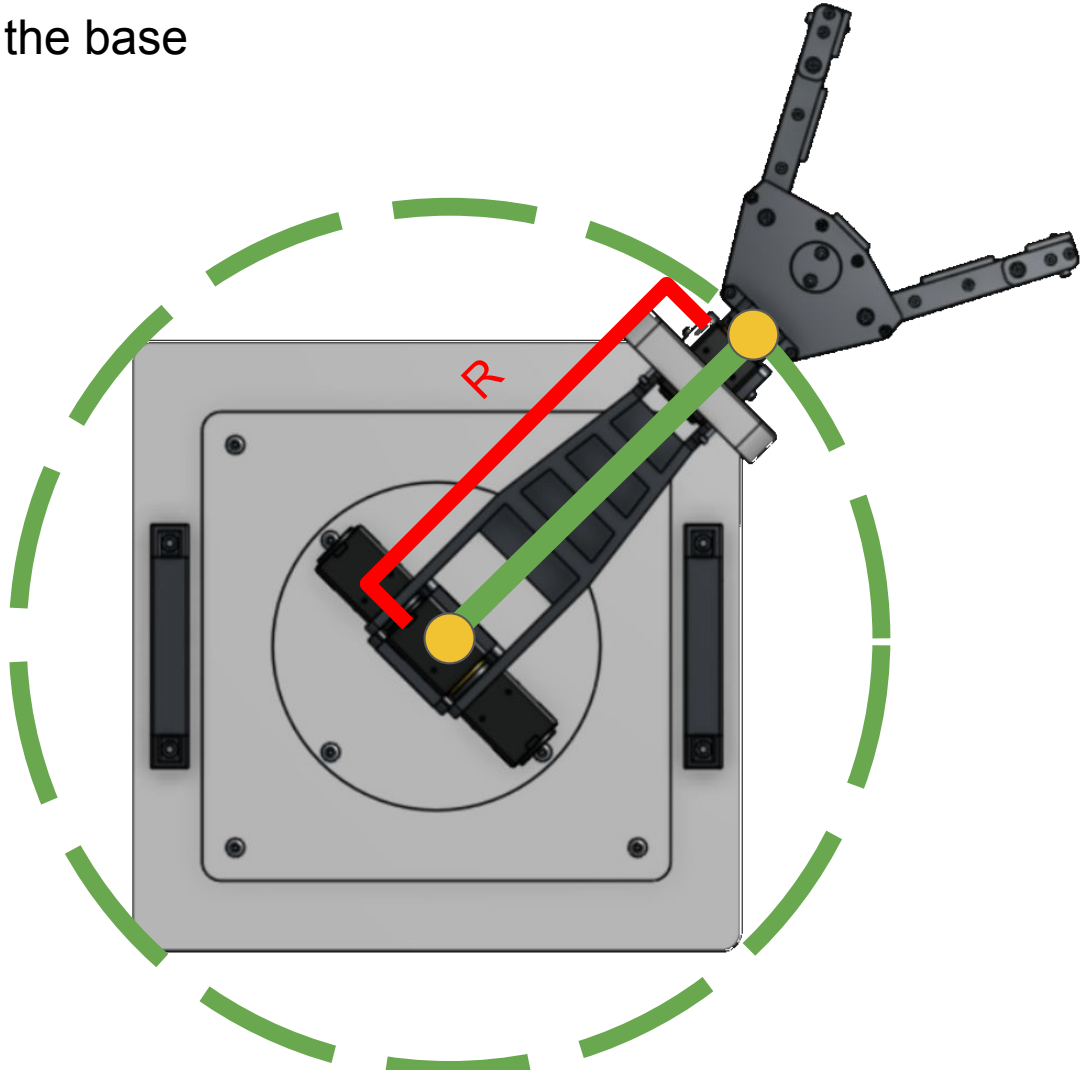
R is variable because the end effector can move away or closer to the base center point, horizontally



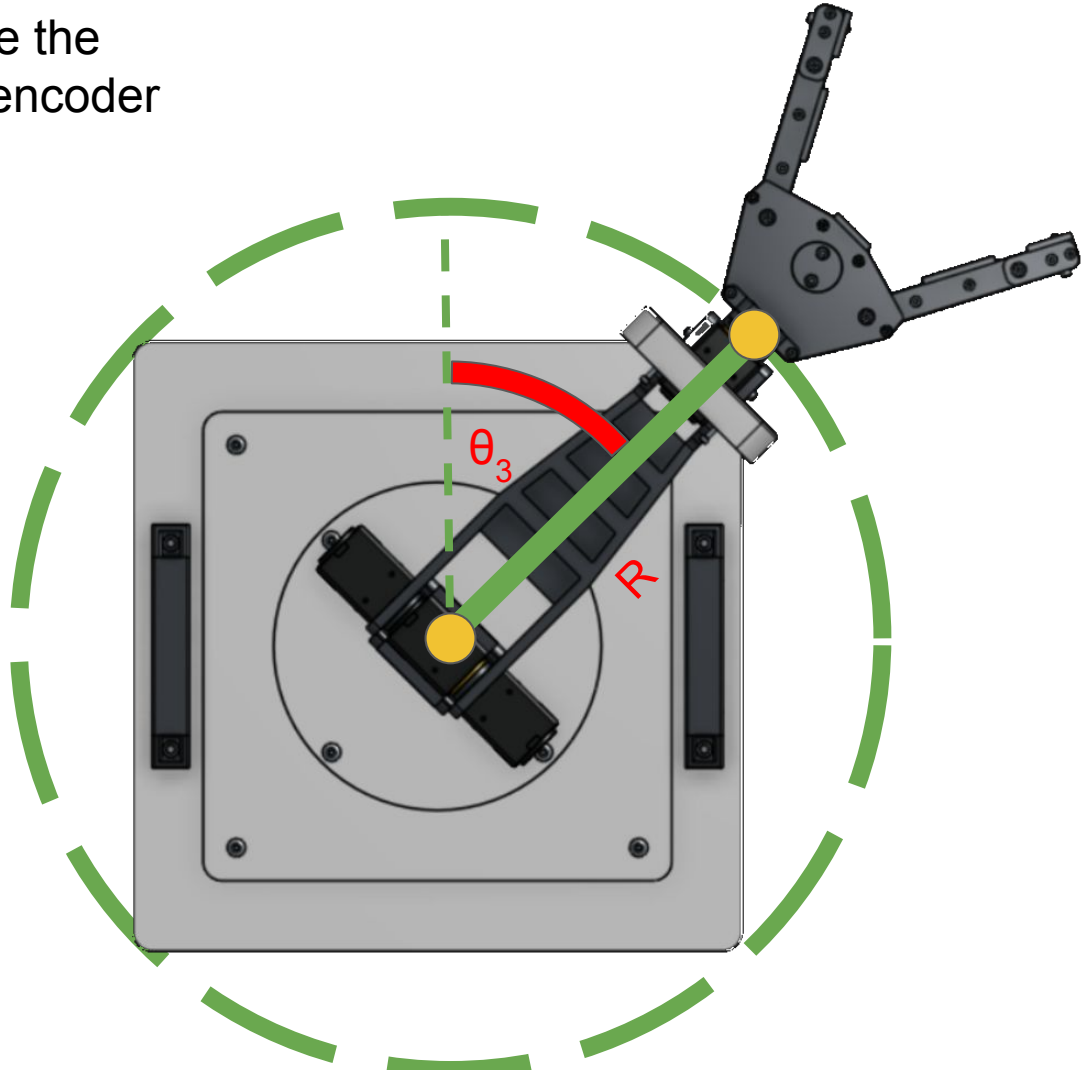
$R_1 < R_2$



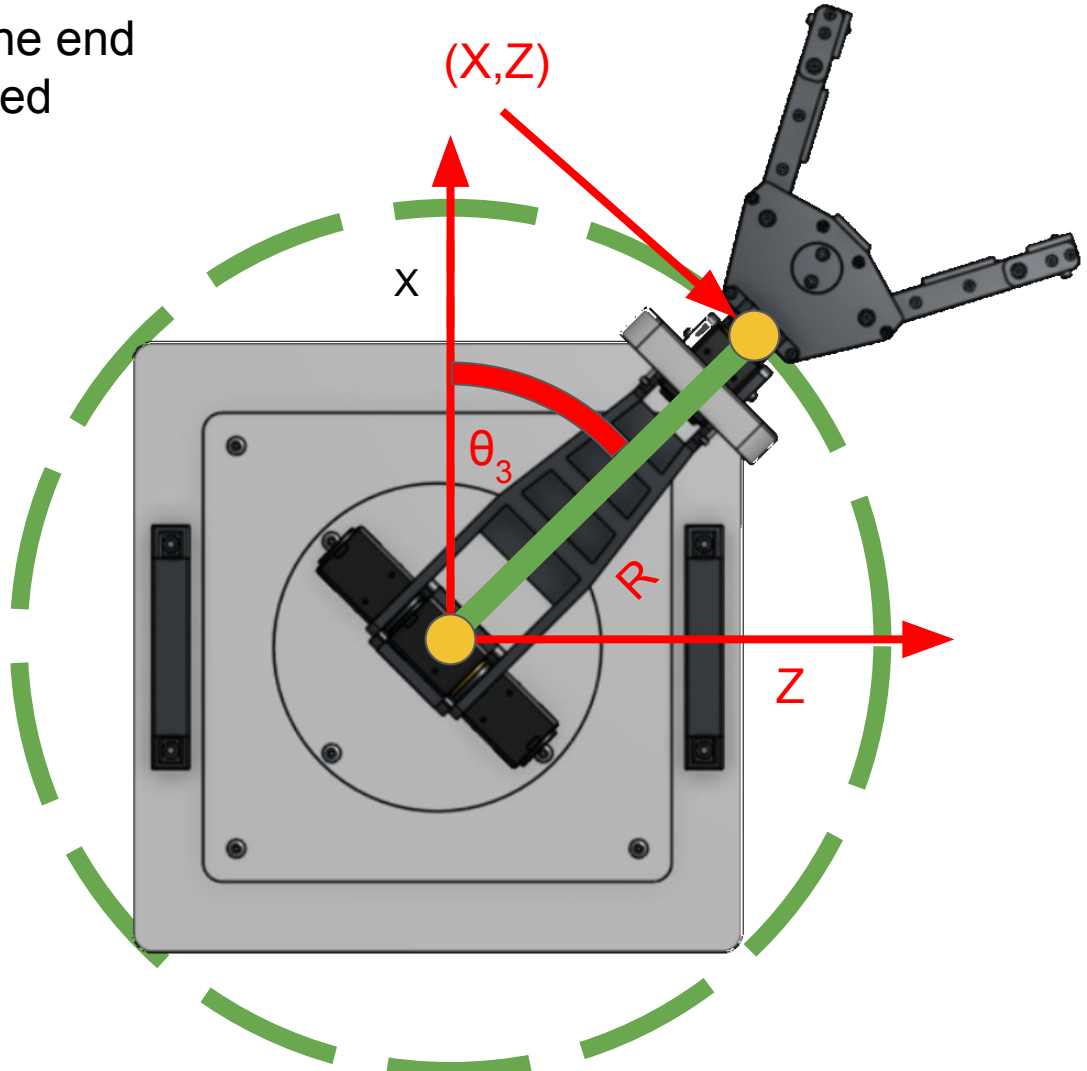
R forms a circle around the center of the base when rotated



θ_3 is known because we can measure the rotation of the base actuator with its encoder

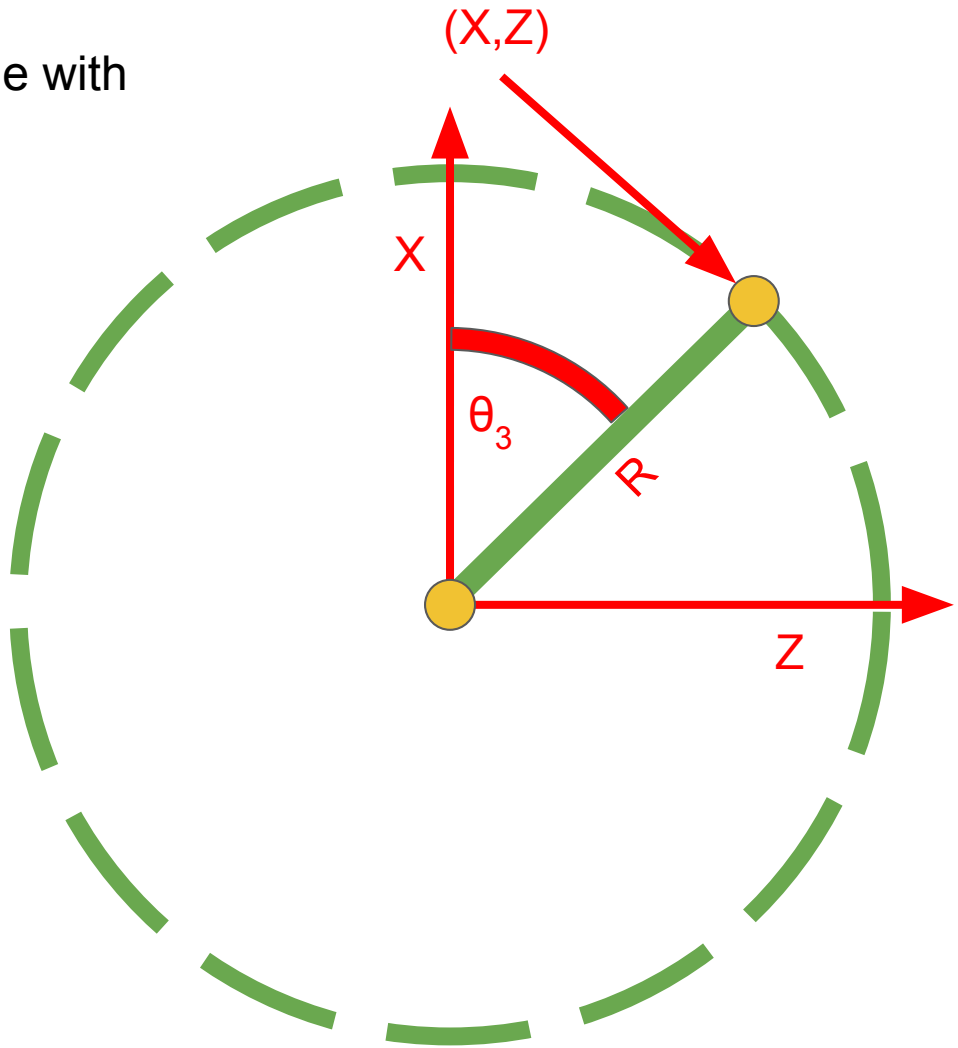


So if R is calculated, the position of the end effector in 3D space can be determined

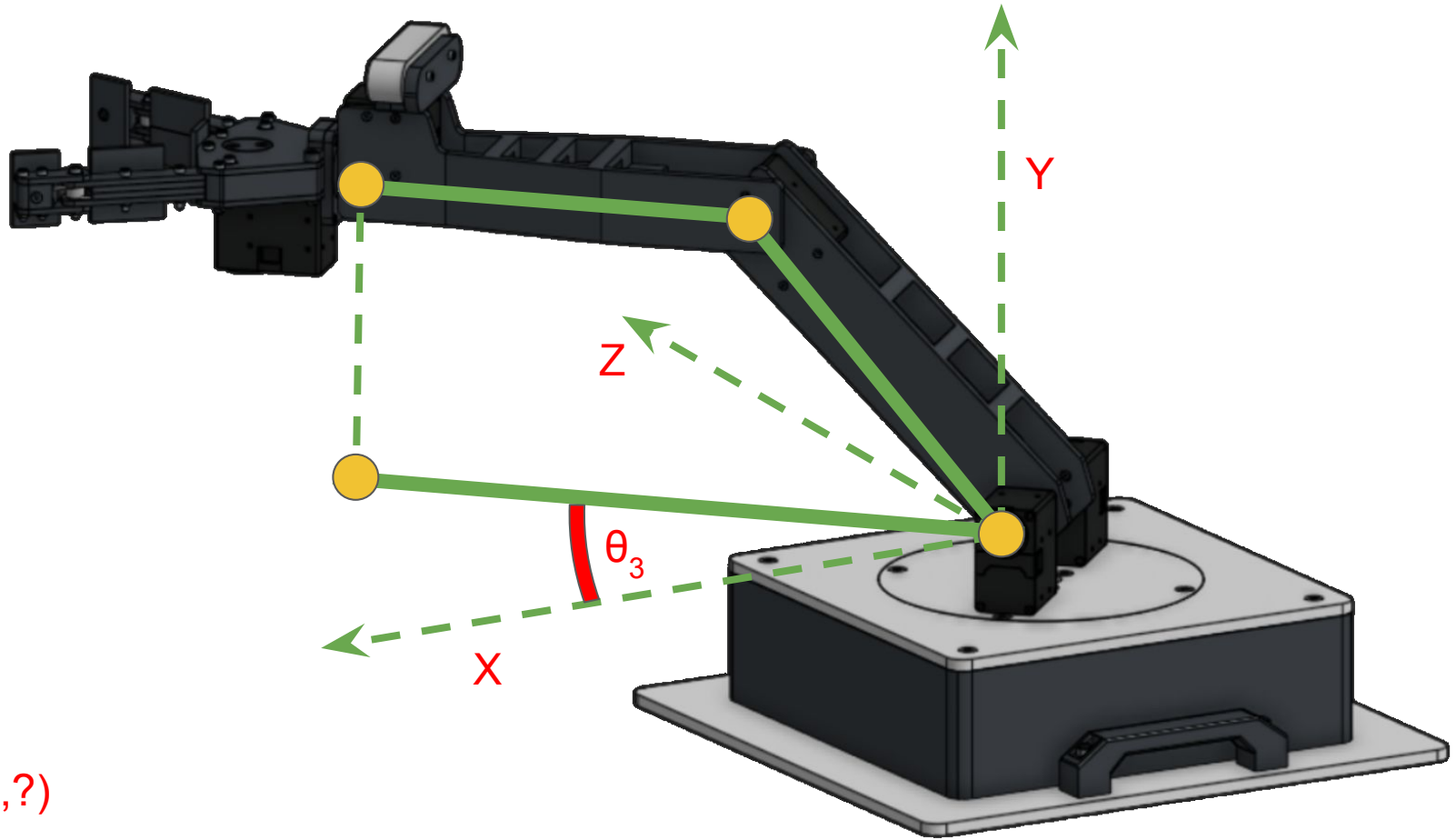


The problem can be represented as a circle with a radius

To find (X,Z)
1. Find R
2. Calculate the location of R on the circle

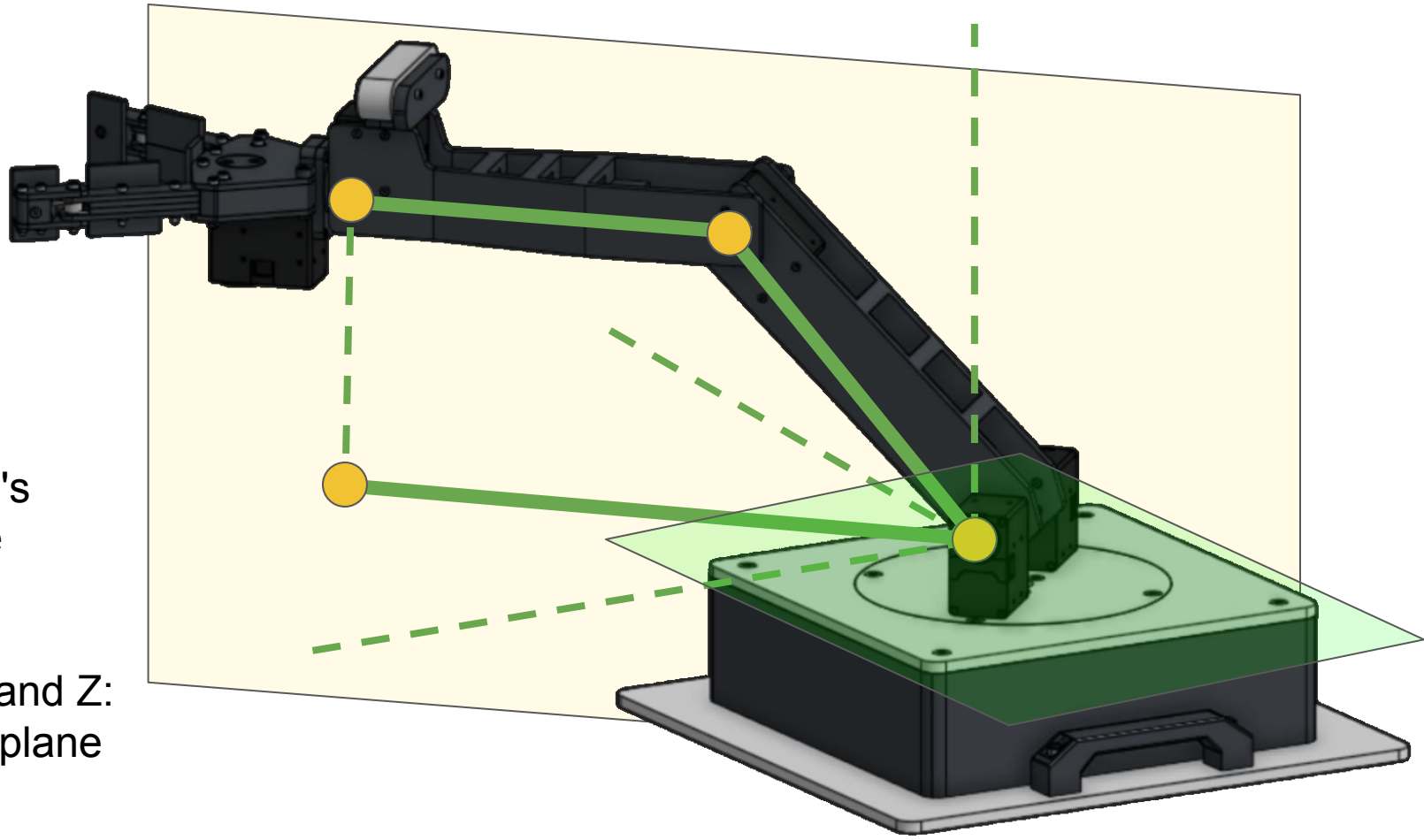


Example Problem



L_1=250mm
L_2=250mm
θ_1=45°
θ_2=45°
θ_3=45°
(X,Y,Z) = (?,?,?)

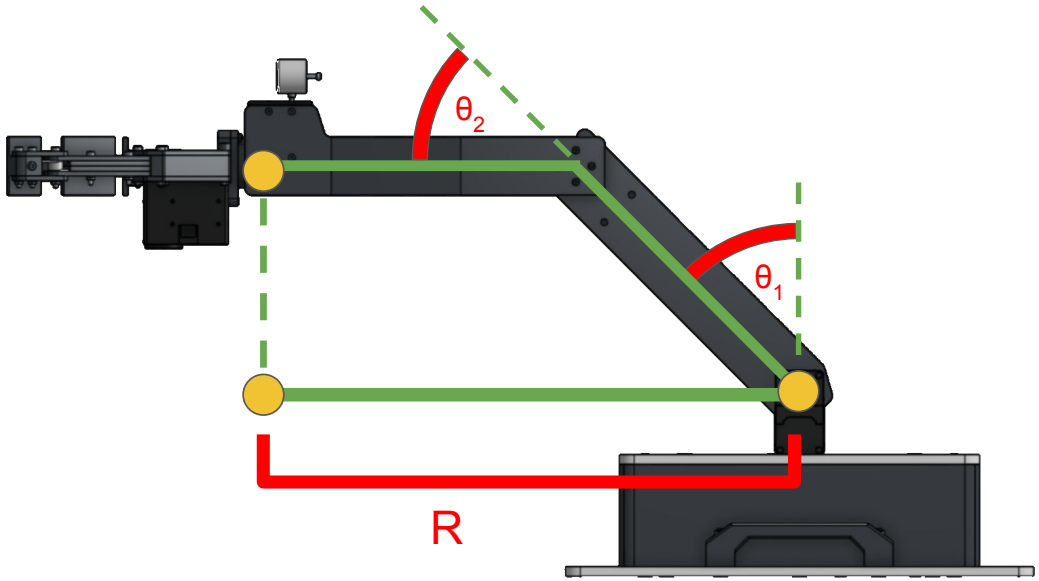
This problem can be split into two 2D problems

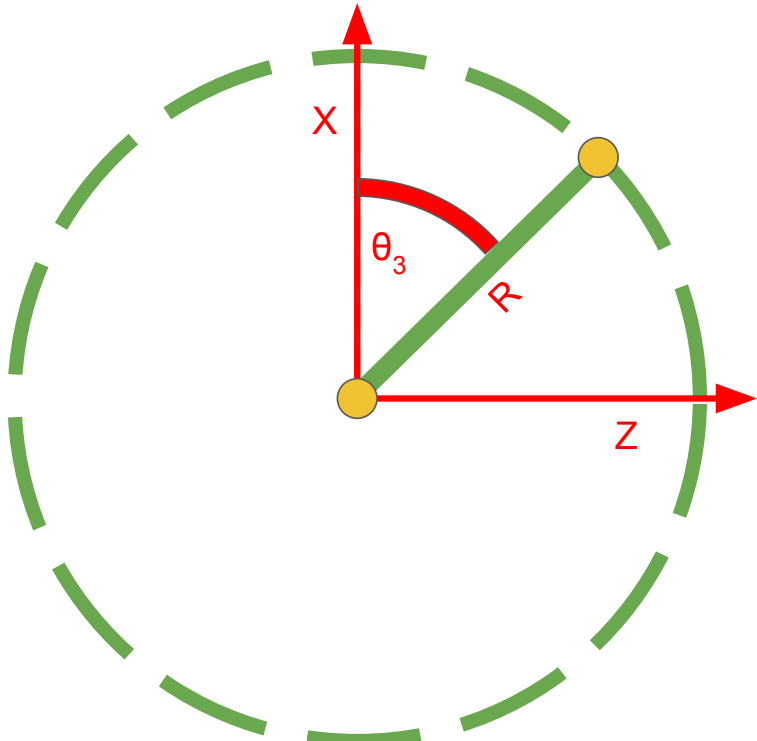


Solving for R: along the arm's own XY plane

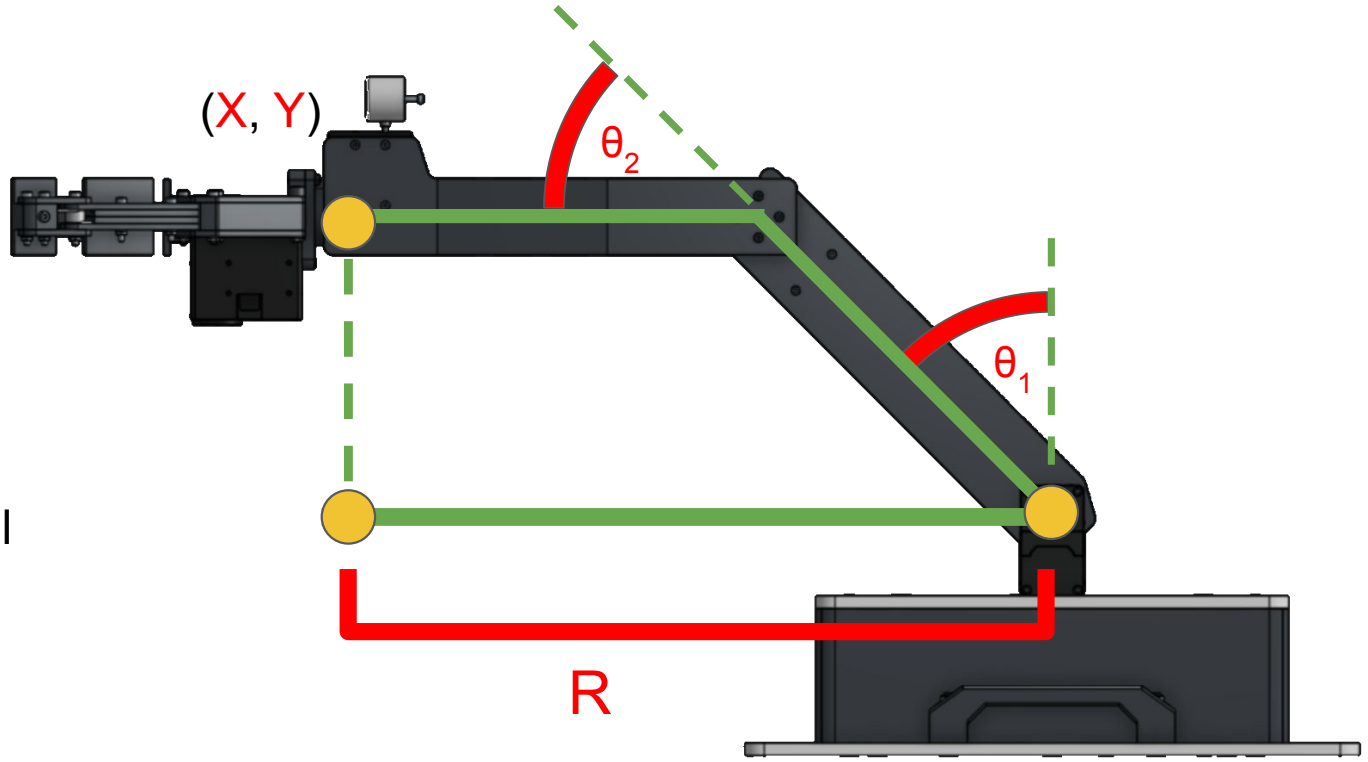
Solving for X and Z: Along the XZ plane

The problem can be solved
along these two planes





R will be equal to the X value we calculated when analyzing this position in 2D



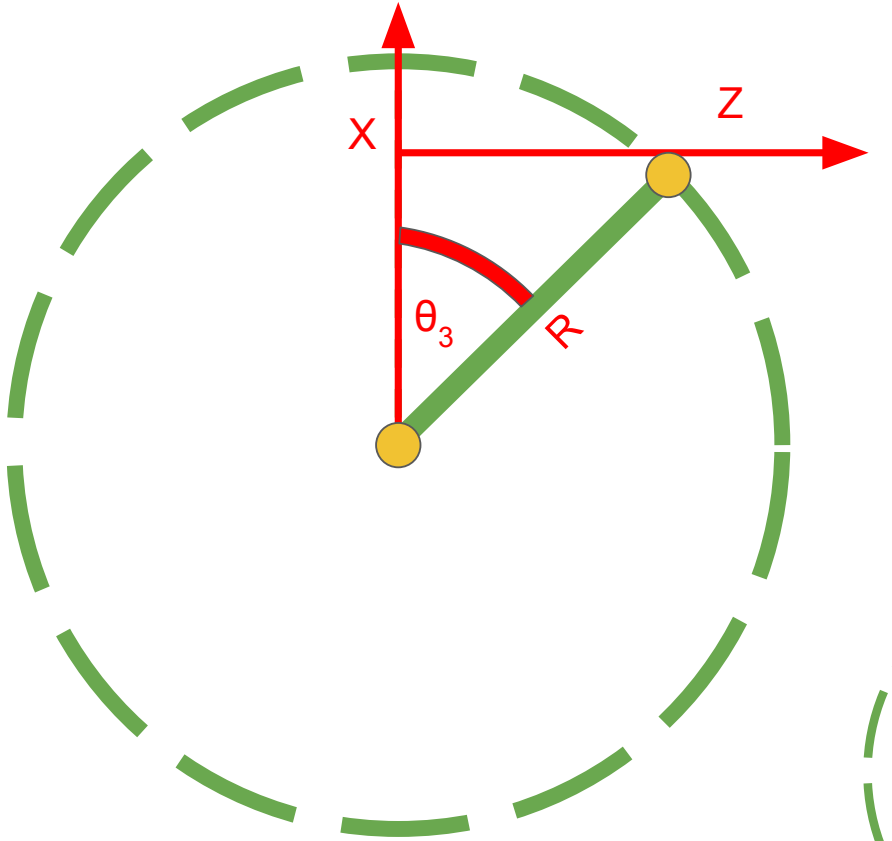
For this problem (L_1=250mm, L_2=250mm, θ_1=45°, θ_2=45°)
We solved it already and determined (X,Y) to equal (426.777, 176.777)
In this frame, R=x
Therefore, R=426.78mm and Y=176.78mm

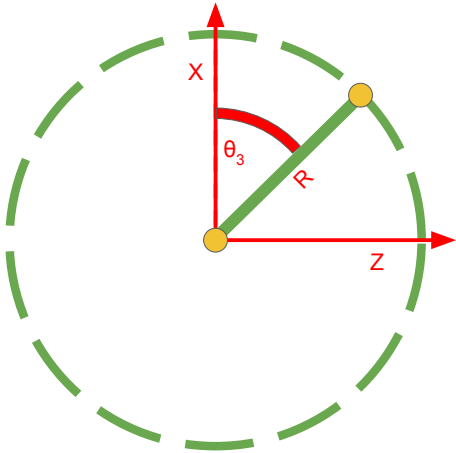
Along the XZ Plane

R= 426.78mm
θ_3= 45°

Z = Rsin(θ_3)
X = Rcos(θ_3)

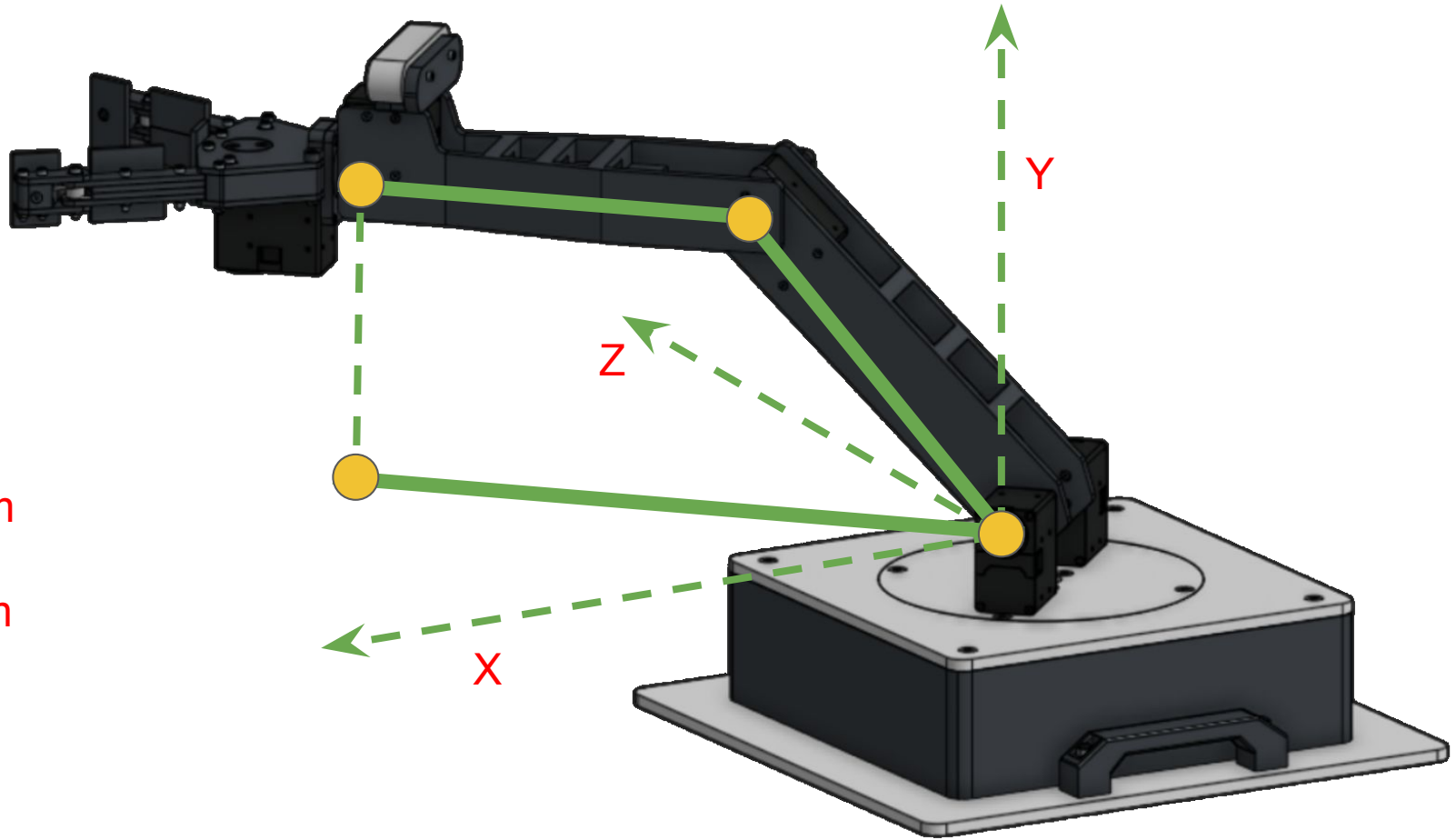
Z = 301.78mm
X = 301.78mm





Example Problem: Solution

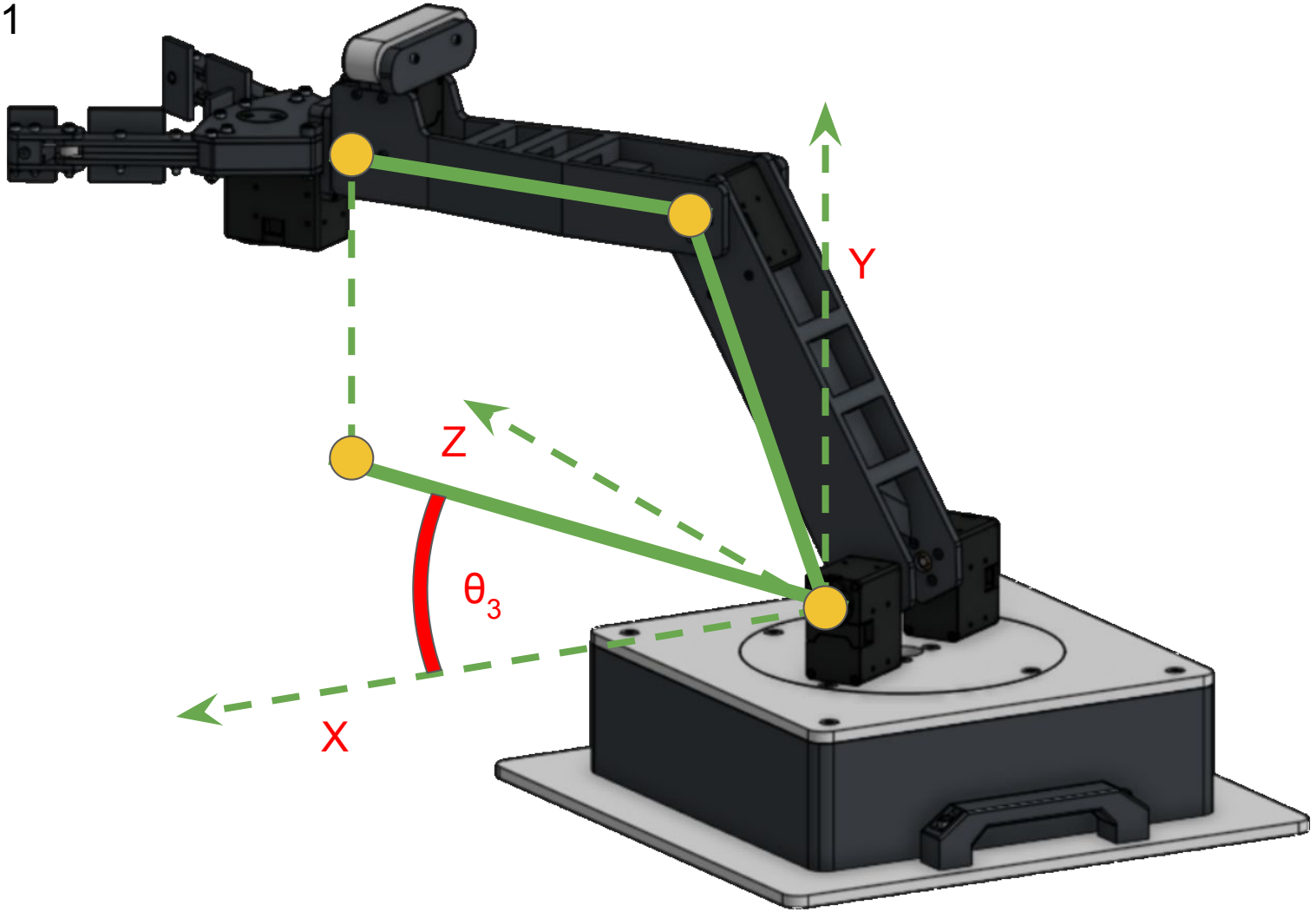
L_1=250mm
L_2=250mm
θ_1=45°
θ_2=45°
θ_3=45°



X = 301.78mm
Y=176.78mm
Z = 301.78mm

Practice Problem 1

L_1=250mm
L_2=250mm
θ_1=30°
θ_2=30°
θ_3=60°
(X,Y,Z) = (?,?,?)

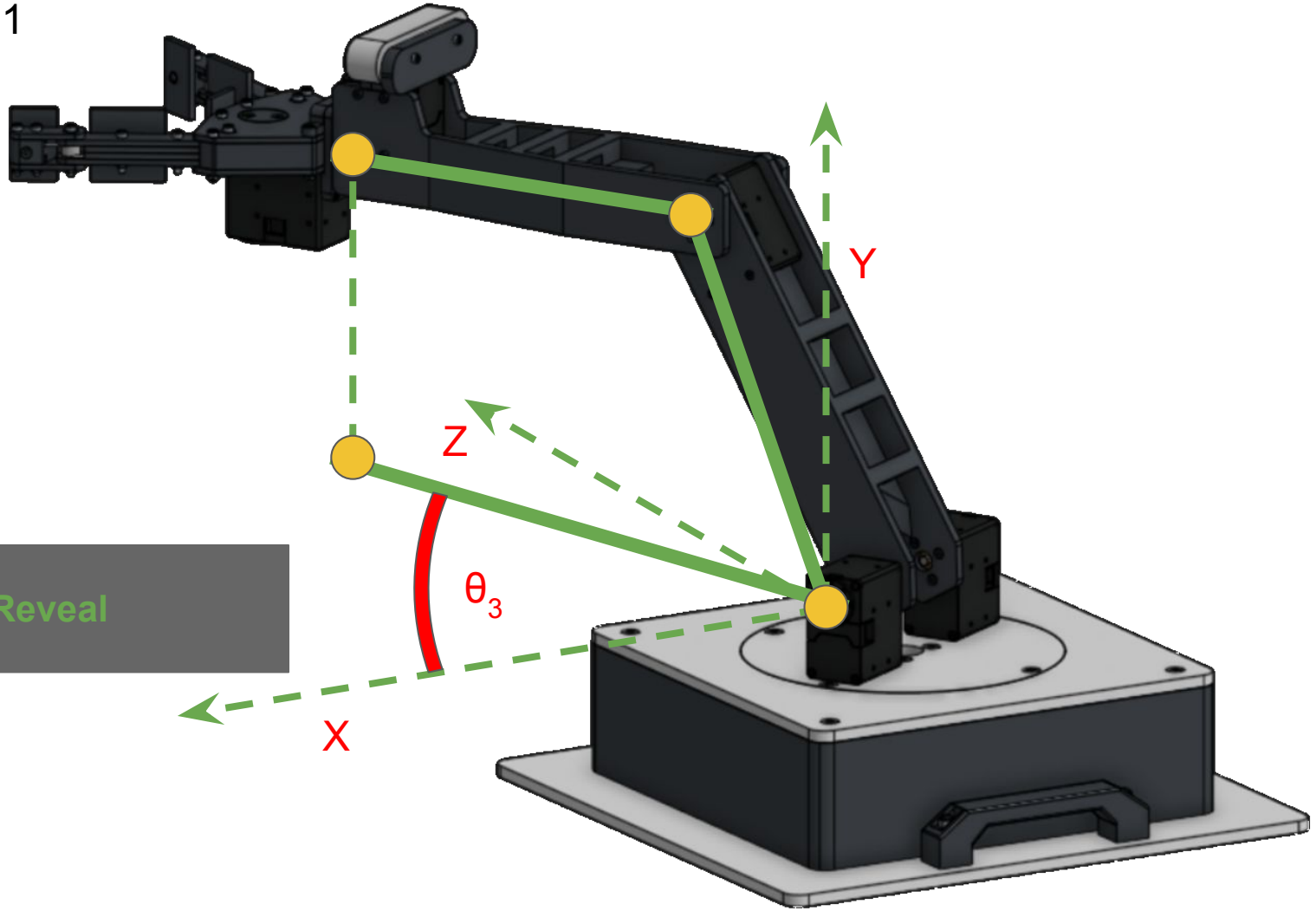


Practice Problem 1

L_1=250mm
L_2=250mm
θ_1=30°
θ_2=30°
θ_3=60°
(X,Y,Z) = (?,?,?)

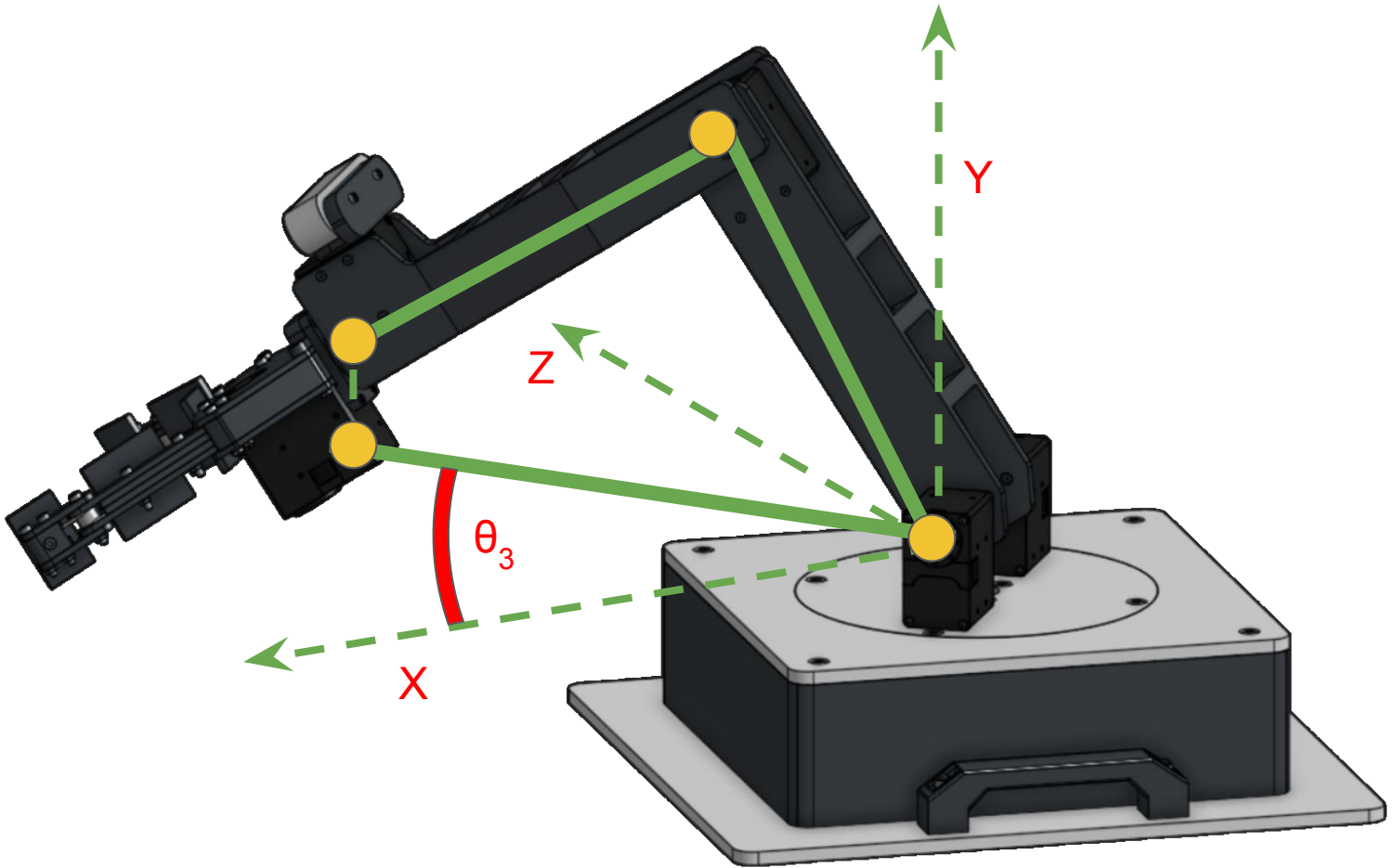
Solution:

Click To Reveal



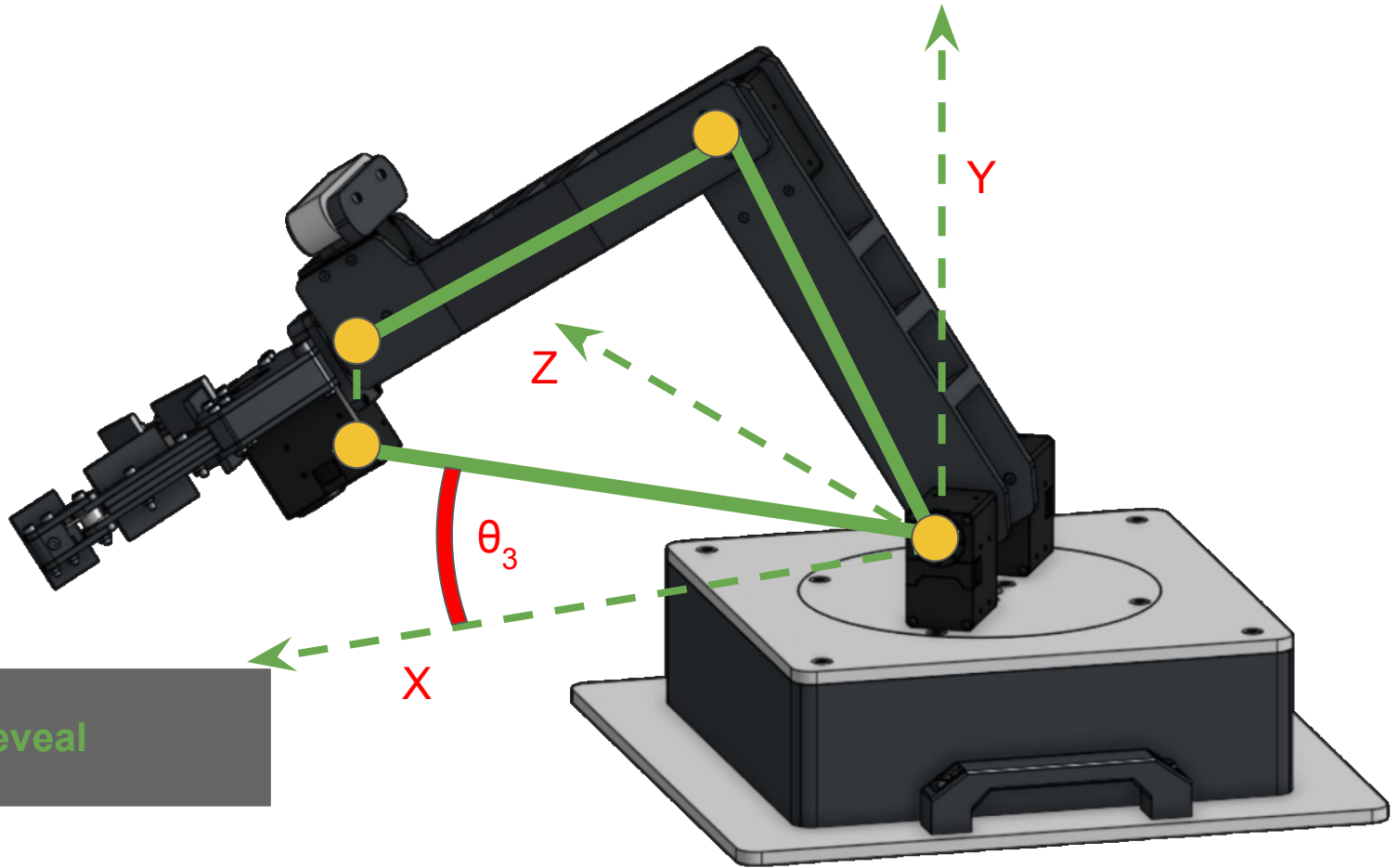
Practice Problem 2

L_1=250mm
L_2=250mm
θ_1=30°
θ_2=90°
θ_3=30°
(X,Y,Z) = (?,?,?)



Practice Problem 2

L_1=250mm
L_2=250mm
θ_1=30°
θ_2=90°
θ_3=30°
(X,Y,Z) = (?,?,?)



Solution:

Click To Reveal

Desmos Calculator

https://www.desmos.com/3d/9bnvqc263x

